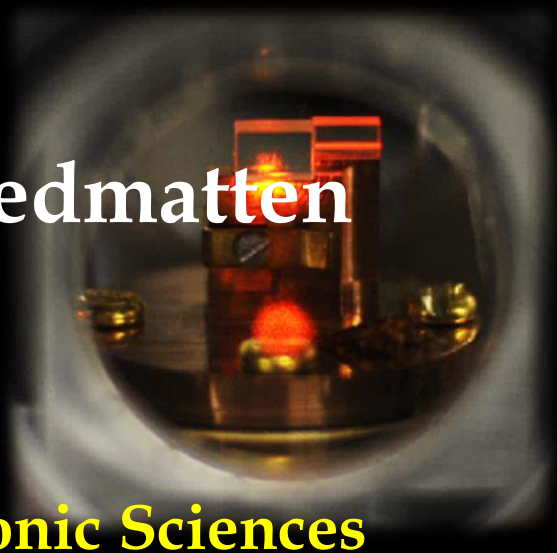
ICFO
Institut de Ciències Fotòniques

Quantum Repeaters





Hugues de Riedmatten

ICFO-The Institute of Photonic Sciences

ICREA- Catalan Institute for Research and Advanced studies

Tutorial, QCRYPT 2015, Tokyo

erc

GOBIERNO DE ESPAÑA
MINISTERIO DE ECONOMÍA Y COMPETITIVIDAD

MARIE CURIE ACTIONS
Marie Curie Initial Training Network
Coherent Information Processing in Rare-Earth Ion Doped Solids

ICFO-The Institute of Photonic Sciences





...Advancing the Frontiers of the Science of Light...

www.icfo.es

Barcelona, Spain

Funded in 2002

Director: Lluis Torner

23 research groups (~280 researchers)

-Quantum Optics

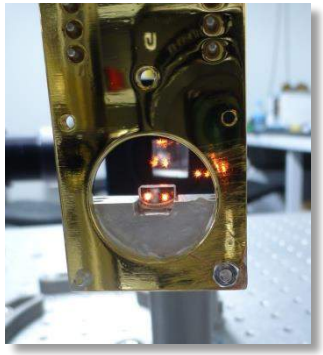
-Nano-Photonics

-Non-Linear Optics

- Bio-Photonics

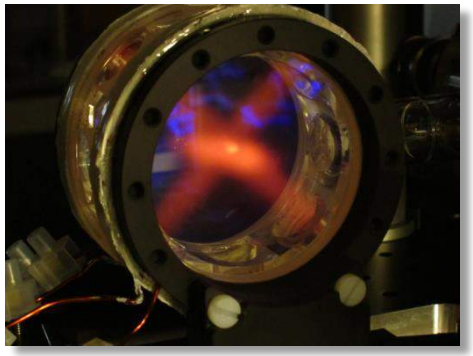
Quantum Photonics @ICFO

Solid State QM



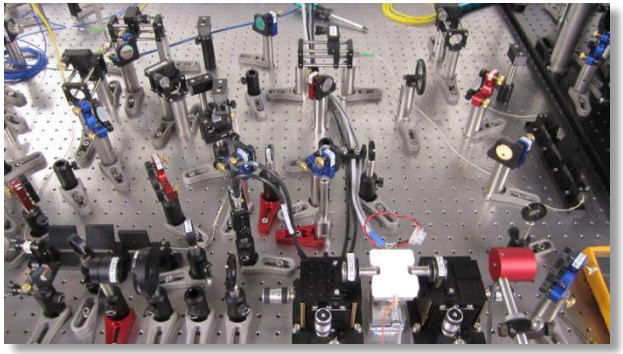
Mustafa Gündogan, Kutlu Kutluer,
Alessandro Seri, Margherita Mazzera

Rb QM & Rydberg ensembles



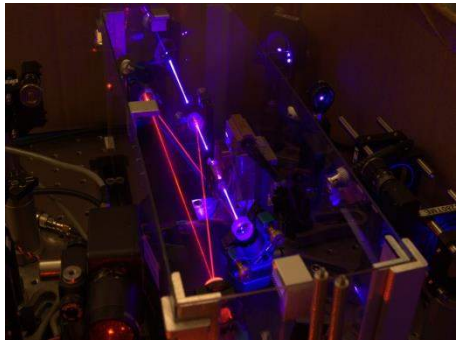
Boris Albrecht, Emanuele Distante
Pau Farrera, Georg Heinze, David Paredes
Auxiliadora Padron

Quantum Frequency conversion



Nicolas Maring, Pau Farrera,
Georg Heinze

Quantum Light sources



Daniel Rieländer, Andreas Lenhard
Margherita Mazzera

Long distance quantum communication

Distribute quantum information (e.g. entanglement) over long distances (continental)

- Quantum key distribution
- Quantum networks
- Long distance tests of quantum physics

Long distance quantum communication

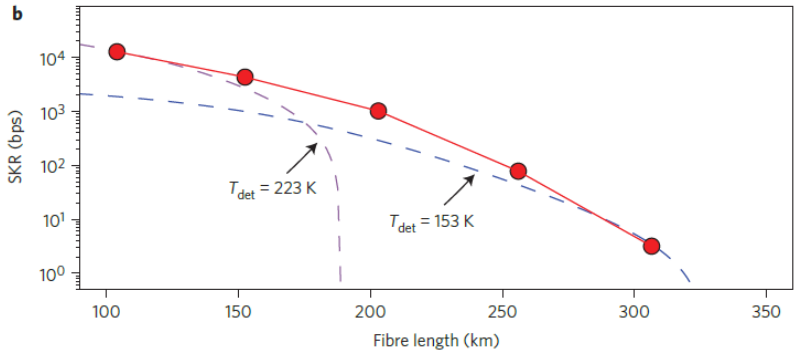
nature photonics

LETTERS

PUBLISHED ONLINE: 9 FEBRUARY 2015 | DOI: 10.1038/NPHOTON.2014.327

Provably secure and practical quantum key distribution over 307 km of optical fibre

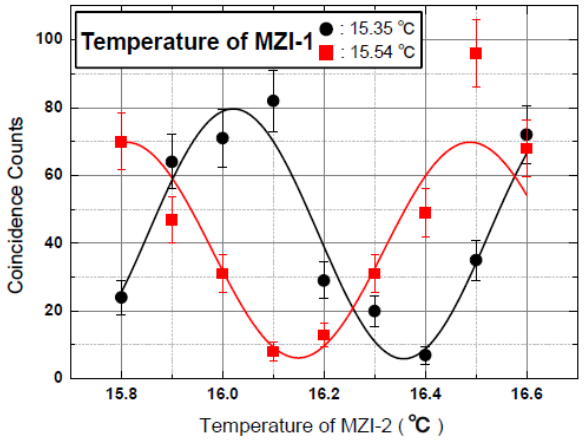
Boris Korzh[1]*, Charles Ci Wen Lim[1]*, Raphael Houlmann[1], Nicolas Gisin[1], Ming Jun Li[2], Daniel Nolan[2], Bruno Sanguinetti[1], Rob Thew[1] and Hugo Zbinden[1]

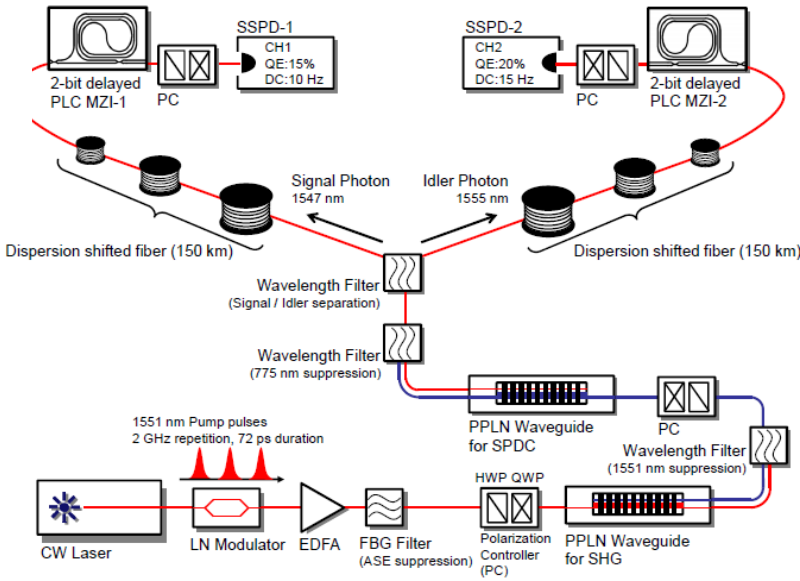


Entanglement distribution over 300 km of fiber

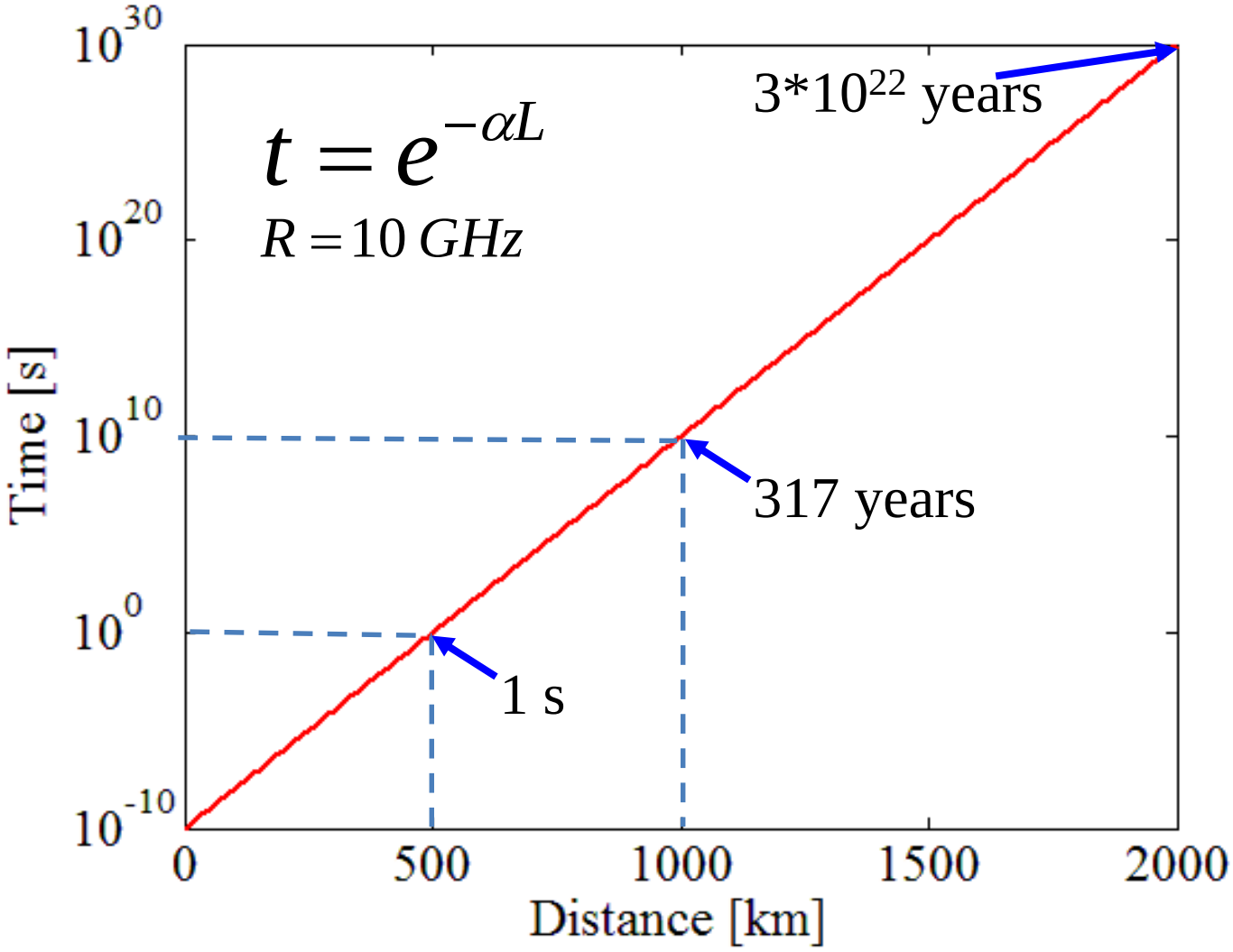
Takahiro Inagaki,[1,*] Nobuyuki Matsuda,[1] Osamu Tadanaga,[2] Masaki Asobe,[2] and Hiroki Takesue[1]

[1]*NTT Basic Research Laboratories, NTT Corporation, Atsugi, Kanagawa, 243-0198, Japan*
[2]*NTT Photonics Laboratories, NTT Corporation, Atsugi, Kanagawa, 243-0198, Japan*
**inagaki.takahiro@lab.ntt.co.jp*





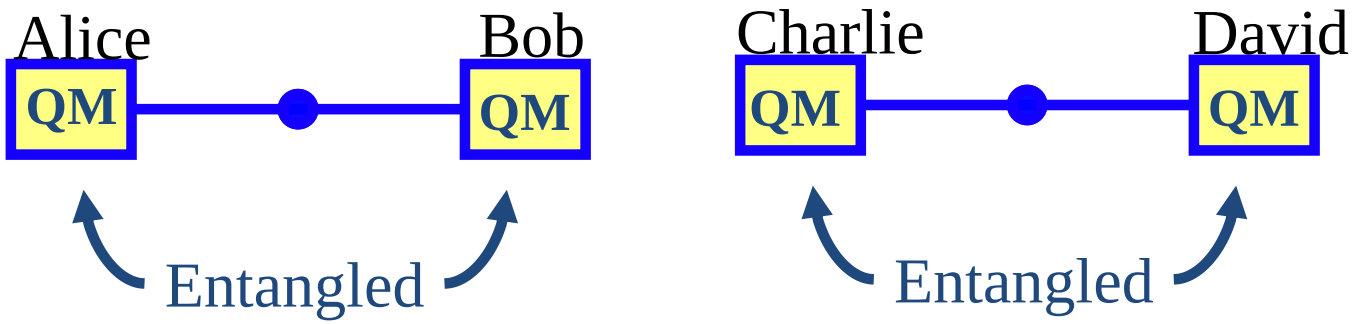
The limits of direct fiber transmission



Outline

- Introduction to quantum repeaters
- Quantum repeaters based on atomic ensembles and linear optics
- Multiplexed quantum repeaters
- Quantum repeaters based on single emitters

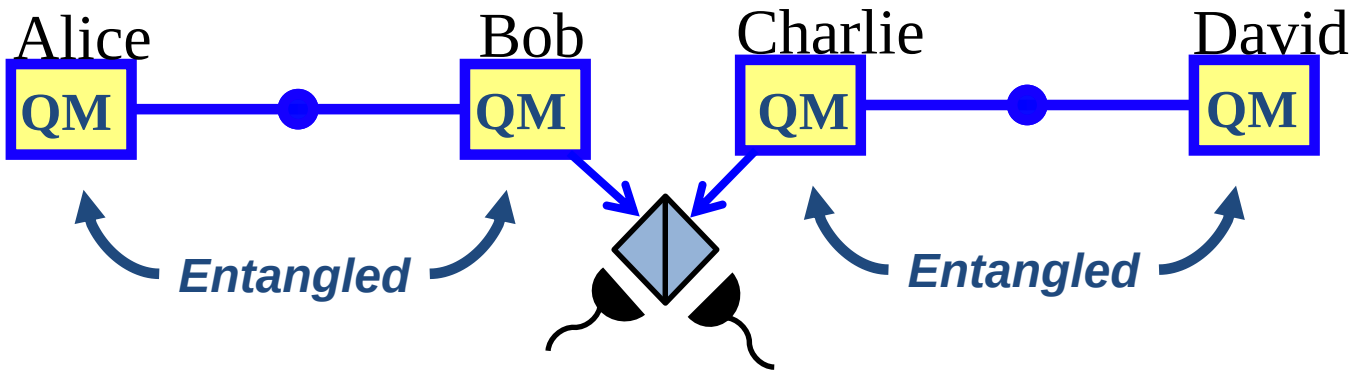
Long distance Quantum Communication with Quantum Repeaters



Create entanglement independently for each link.

H.J. Briegel W.Dur, J.I. Cirac, P.Zoller, PRL **81**, 5932 (1998)
L.M. Duan, M.D. Lukin, J.I. Cirac, P.Zoller, Nature **414**, 413 (2001)
N. Sangouard, C. Simon, H. de Riedmatten and N. Gisin, Rev. Mod. Phys. **83**, 33–80 (2011)

Long distance Quantum Communication with Quantum Repeaters

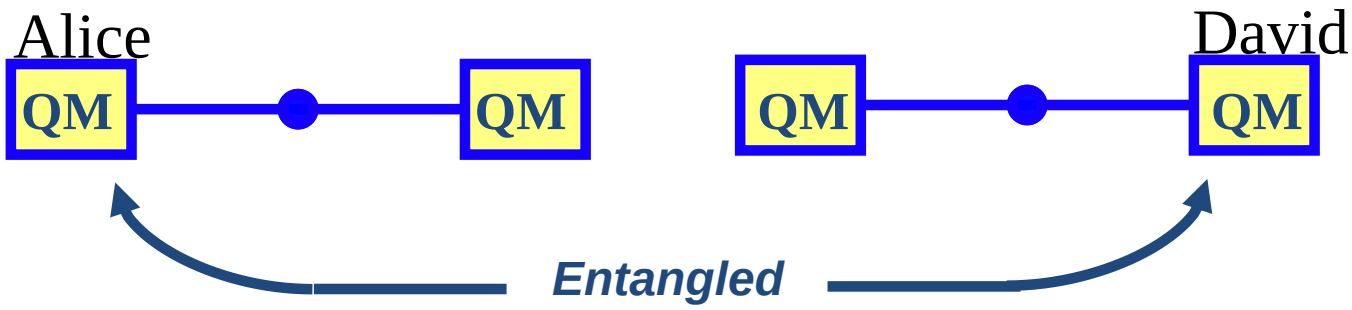


Create entanglement independently for each link.

Extend by entanglement swapping.

H.J. Briegel W.Dur, J.I. Cirac, P.Zoller, PRL **81**, 5932 (1998)
L.M. Duan, M.D. Lukin, J.I. Cirac, P.Zoller, Nature **414**, 413 (2001)
N. Sangouard, C. Simon, H. de Riedmatten and N. Gisin, Rev. Mod. Phys. **83**, 33–80 (2011)

Long distance Quantum Communication with Quantum Repeaters



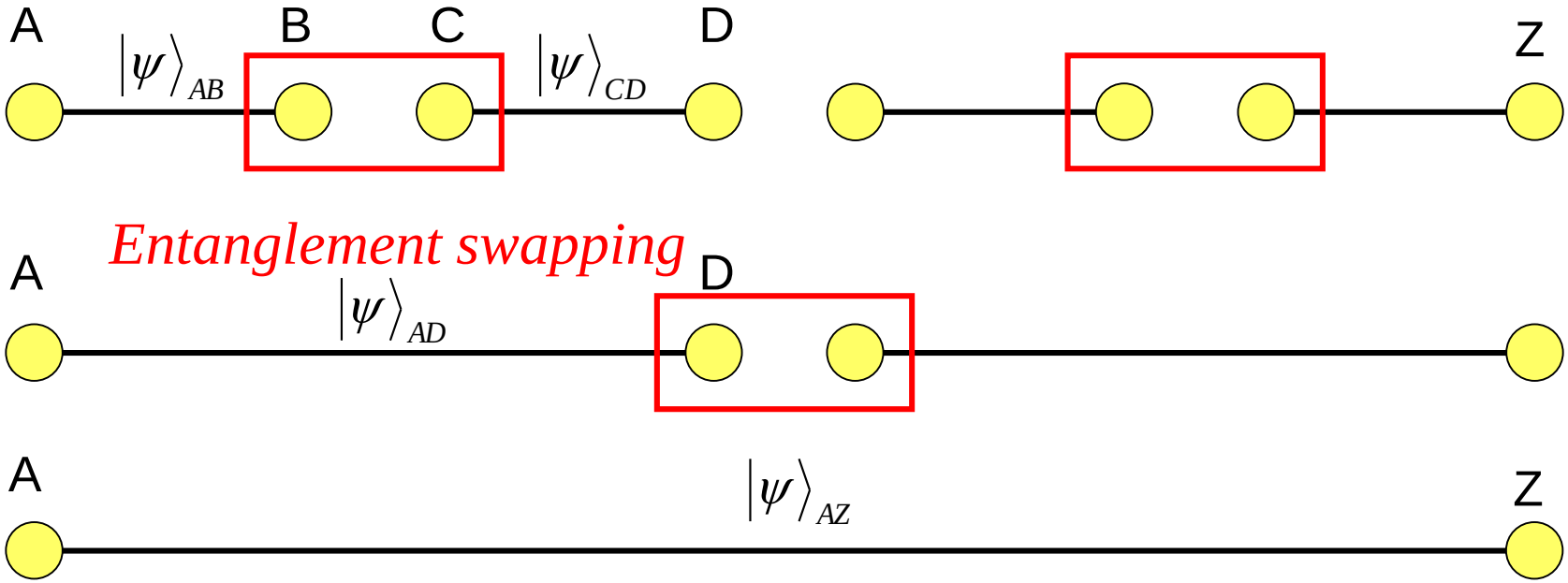
Create entanglement independently for each link.

Extend by entanglement swapping.

Requires heralded creation and storage of entanglement

H.J. Briegel W.Dur, J.I. Cirac, P.Zoller, PRL **81**, 5932 (1998)
L.M. Duan, M.D. Lukin, J.I. Cirac, P.Zoller, Nature **414**, 413 (2001)
N. Sangouard, C. Simon, H. de Riedmatten and N. Gisin, Rev. Mod. Phys. **83**, 33–80 (2011)

Long distance Q Comm with Quantum Repeaters



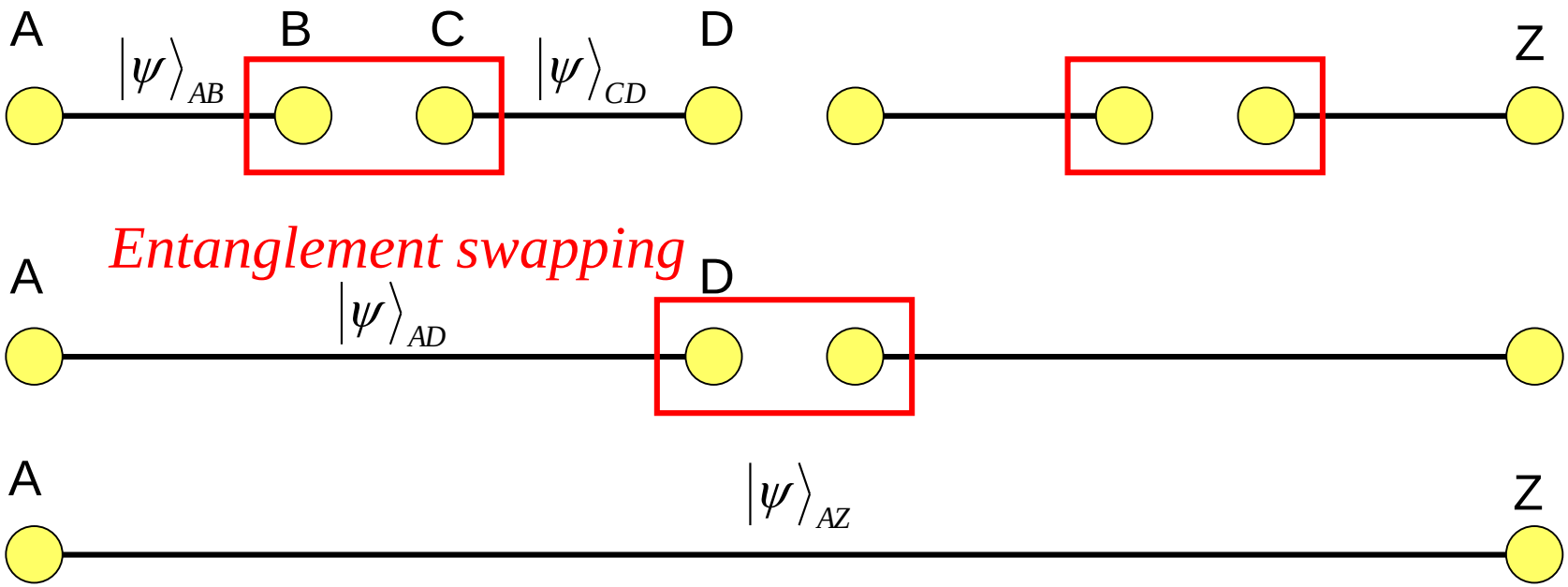
$$T_{tot} \approx \left(\frac{3}{2}\right)^n \frac{L_0}{c} \frac{1}{P_0 P_1 \cdots P_n}$$

H.J. Briegel W.Dur, J.I. Cirac, P.Zoller, PRL 81, 5932 (1998)

L.M. Duan, M.D. Lukin, J.I. Cirac, P.Zoller, Nature 414, 413 (2001)

C. Simon, H. de Riedmatten, M. Afzelius, N. Sangouard, H. Zbinden and N. Gisin, PRL 98, 190503 (2007)

Long distance Q Comm with Quantum Repeaters



Other new protocols not based on heralded entanglement and quantum memories. Require more advanced capabilities (see next talk by Liang Jiang)

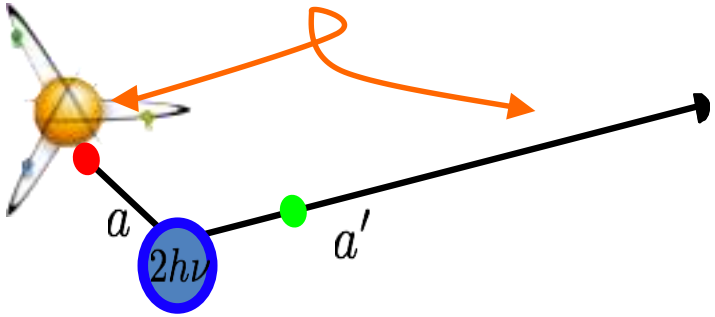
H.J. Briegel W.Dur, J.I. Cirac, P.Zoller, PRL 81, 5932 (1998)

L.M. Duan, M.D. Lukin, J.I. Cirac, P.Zoller, Nature 414, 413 (2001)

C. Simon, H. de Riedmatten, M. Afzelius, N. Sangouard, H. Zbinden and N. Gisin, PRL 98, 190503 (2007)

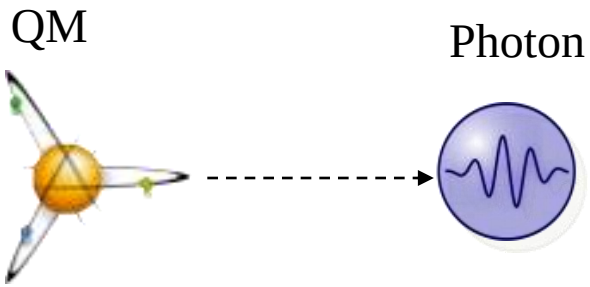
A crucial resource : Light-matter quantum entanglement

Read-write Quantum memory
+quantum light source
(e.g. Entanglement source)



☺ Wavelength flexibility.
Possibility to create entanglement between photon at telecom wavelength and QM

Read quantum memory



☺ Source and memory in one system
☹ No wavelength flexibility
Emission not at telecom wavelengths

Quantum light matter interfaces

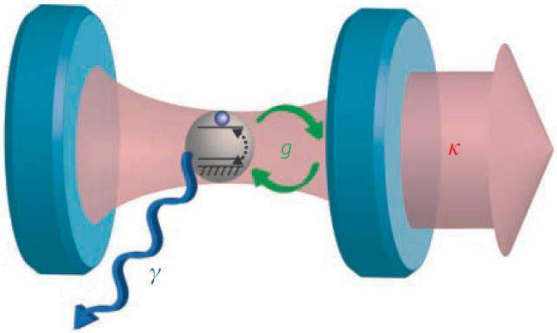
- Single quantum systems (single atoms)



Very small interaction probability between single photons and single atoms in free space (under standard conditions):

$$P_{sc} \propto \frac{\lambda^2}{A}$$

- Solution I : Put the atom in a cavity : cavity QED



H.J. Kimble, Nature (2008)

$$P_{sc} \propto \frac{\lambda^2}{A} N_{bounces}$$

Need high cooperativity $C = \frac{g^2}{\kappa\gamma}$

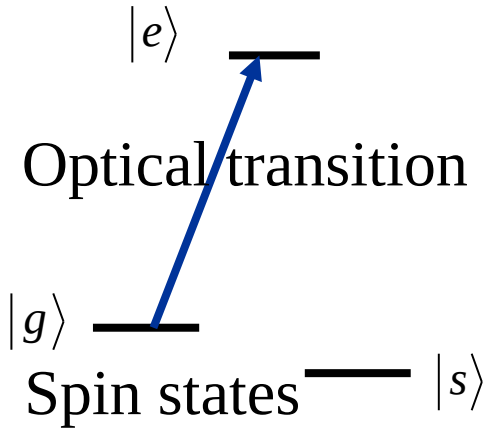
Strong Coupling $g >> (\gamma, \kappa)$

Ideal system, but complicated to implement ! (see experiment by Kimble, Rempe)

Quantum memories using atomic ensembles

$$P_{sc} \propto \frac{\lambda^2}{A} N_{atoms}$$





- Strong light-matter coupling without cavities $\quad g \rightarrow g\sqrt{N_{atoms}}$

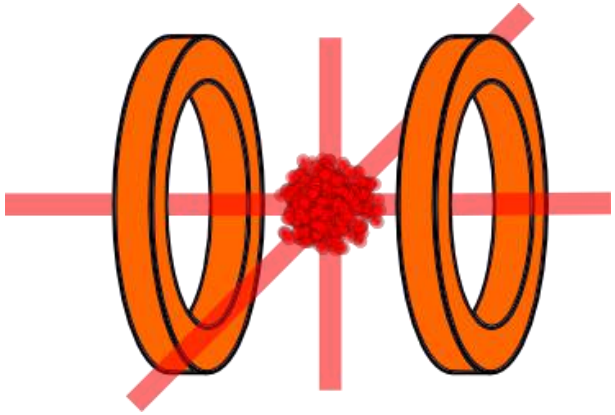
- Qubits encoded in collective excitations : Efficient Collective retrieval

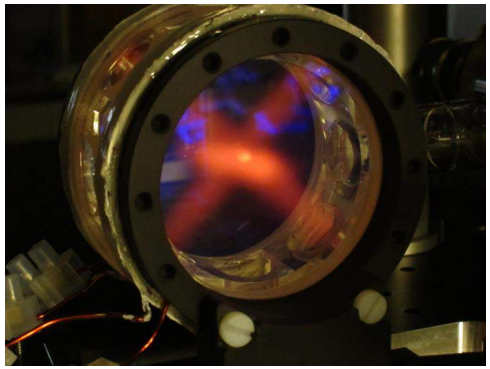
$$\left|\Psi_{coll}\right\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{i\vec{k}\vec{r}_j} \left| g_1 g_2 .. e_j .... g_N \right\rangle$$

- Quantum Info multiplexing

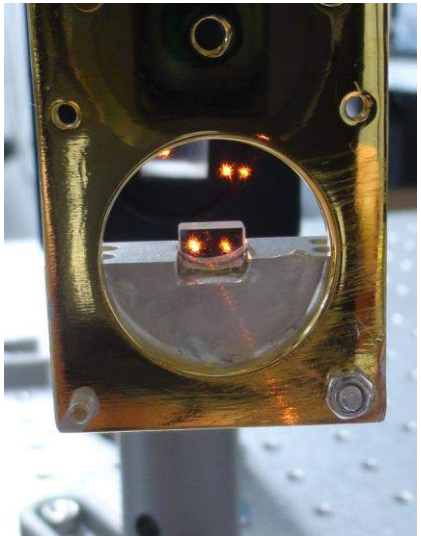
Ensemble based quantum memories

Ensembles of
laser cooled atoms
Magneto-Optical Trap (e.g. Rb, Cs)





Solid state atomic ensembles
based on
Rare-earth doped solids



Ensemble based quantum memories

Atomic gases

Atomic ensemble in Solid State
Rare-earth ion doped crystals



Far-off resonance Raman

DLCZ

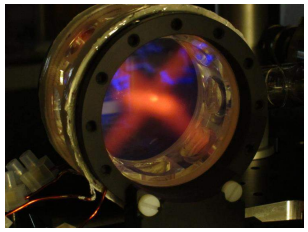
Electro-magnetically induced Transparency

Photon echo based protocols
GEM, Atomic Frequency Comb

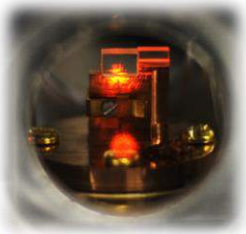
Outline

- Introduction to quantum repeaters

- Quantum repeaters based on atomic ensembles and linear optics



  - The DLCZ protocol
  - Coupling cold atomic QMs to telecom wavelengths
  - Photon pair source and solid-state quantum memories
  - More deterministic schemes



- Multiplexed quantum repeaters

- Quantum repeaters based on single emitters
  - Quantum repeaters based on trapped ions
  - Cavity-based single atoms quantum memories

The DLCZ protocol

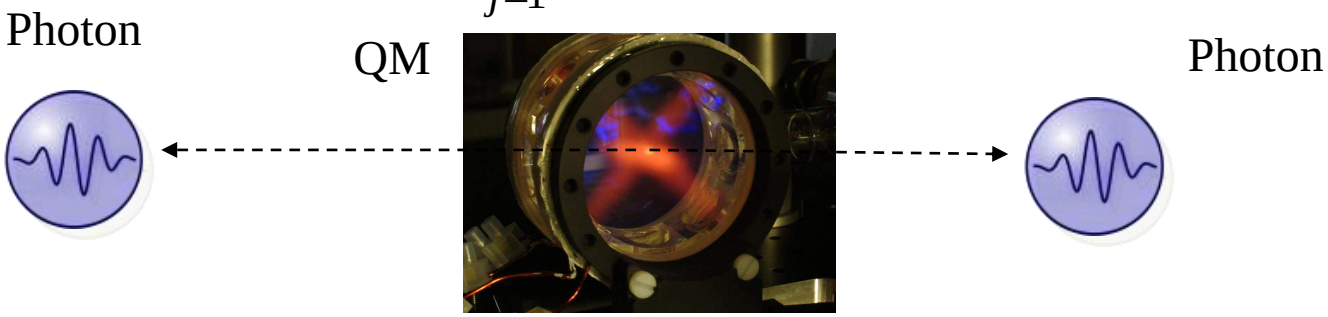
Long-distance quantum communication with atomic ensembles and linear optics

L.-M. Duan*†, M. D. Lukin‡, J. I. Cirac* & P. Zoller*

Nature 414, 413 (2001).

Quantum repeater protocol based on the creation of single collective spin excitations in laser cooled atomic gases using classical pulses. Spin excitation entangled with emitted photon

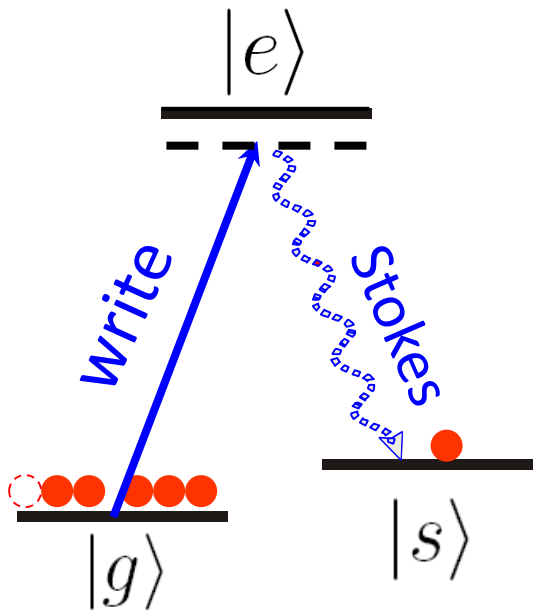
$$|1\rangle_A = \sum_{j=1}^{N} |g_1 \ldots s_j \ldots g_N\rangle$$



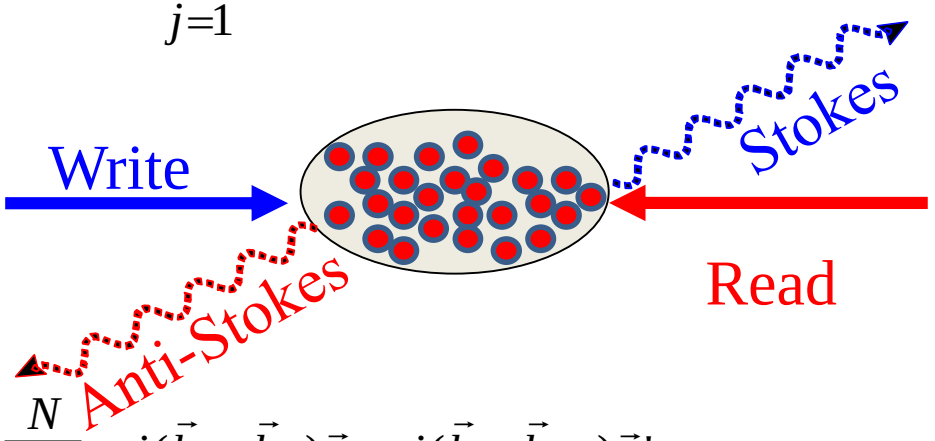
The spin excitation can be efficiently transferred to single photon fields in a well defined direction and at well defined time

Correlated Photon Pairs with a controllable delay

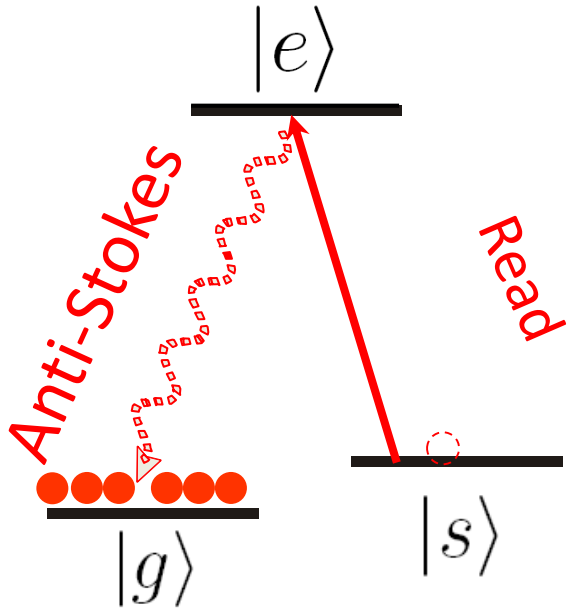
Single collective spin excitations



$$|1\rangle_A = \sum_{j=1}^{N} e^{i(\vec{k}_w - \vec{k}_S)\vec{x}_j} \left| g_1 \ldots . s_j \ldots g_N \right\rangle$$



$$\sum_{j=1}^{N} e^{i(\vec{k}_w - \vec{k}_S)\vec{x}_j} e^{i(\vec{k}_r - \vec{k}_{AS})\vec{x}'_j} \left| g_1 \ldots . g_N \right\rangle$$



Phase matching for collective interference

Atoms at rest: $\vec{k}_w + \vec{k}_r = \vec{k}_S + \vec{k}_{AS}$

Atoms moving $\vec{k}_w = \vec{k}_S, \vec{k}_r = \vec{k}_{AS}$

DLCZ quantum memories : state of the art

ARTICLES
PUBLISHED ONLINE: 26 SEPTEMBER 2010 | DOI: 10.1038/NPHYS1773
nature physics

A quantum memory with telecom-wavelength conversion

A. G. Radnaev†, Y. O. Dudin†, R. Zhao, H. H. Jen, S. D. Jenkins, A. Kuzmich and T. A. B. Kennedy*

100 ms storage time
But low retrieval efficiency

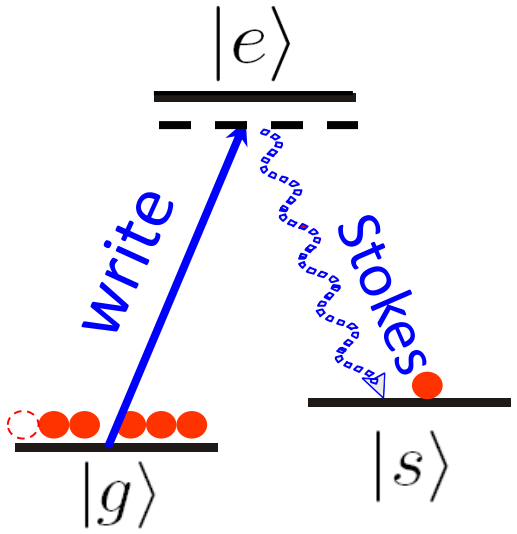
nature physics
LETTERS
PUBLISHED ONLINE: 20 MAY 2012 | DOI: 10.1038/NPHYS2324

Efficient and long-lived quantum memory with cold atoms inside a ring cavity

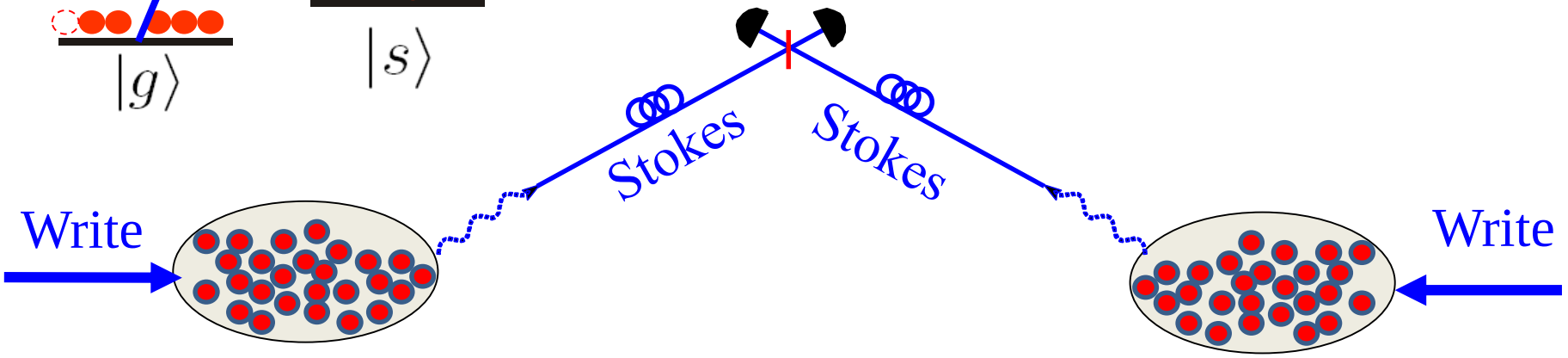
Xiao-Hui Bao[1,2], Andreas Reingruber[1], Peter Dietrich[1], Jun Rui[2], Alexander Dück[1], Thorsten Strassel[1], Li Li[2], Nai-Le Liu[2], Bo Zhao[2]* and Jian-Wei Pan[1,2]*

Combined high efficiency (80 %) and ms storage time

How to create entanglement between remote quantum memories ?



Entanglement at a distance by measurement

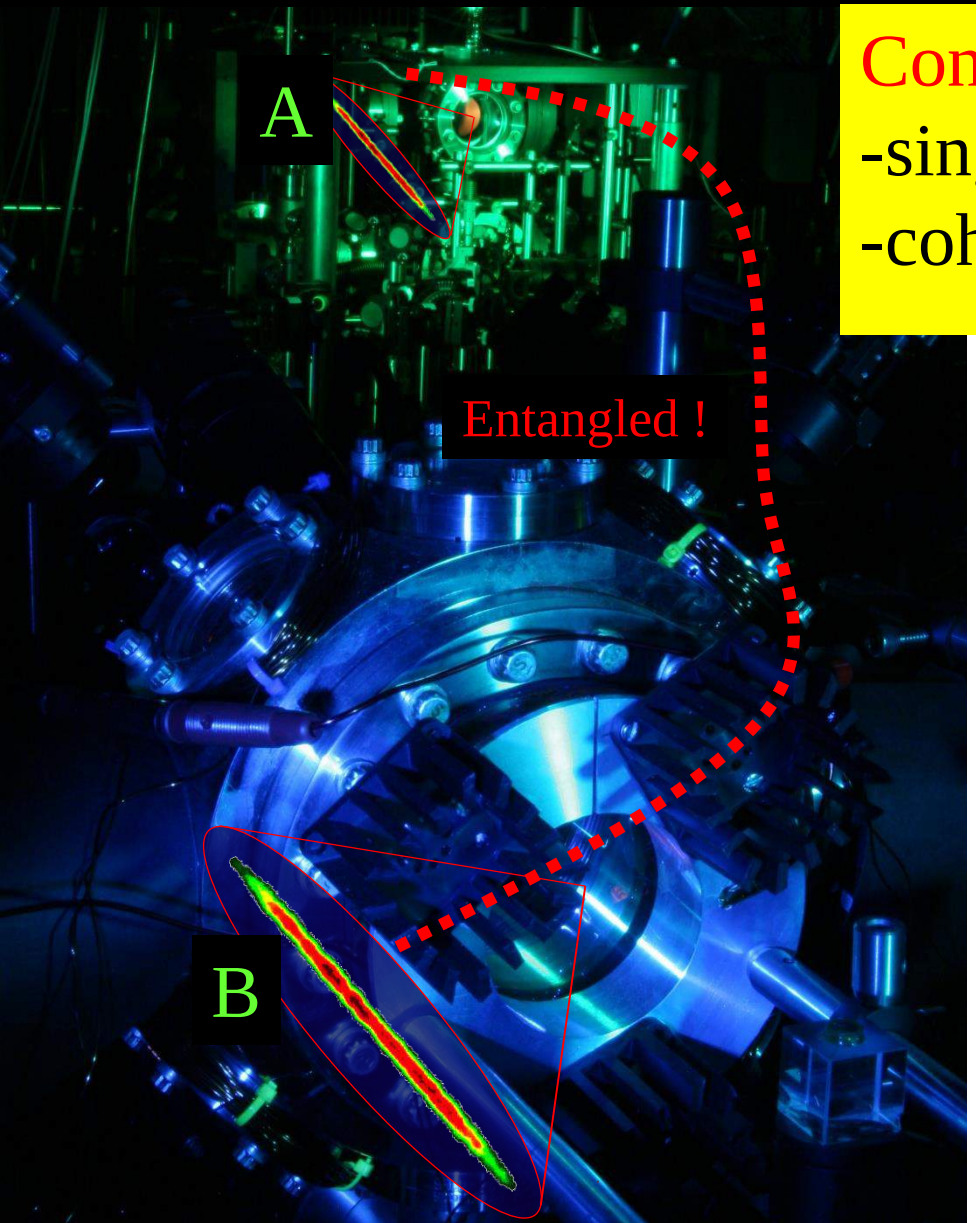


$$|\Phi_{AB}\rangle = \frac{1}{\sqrt{2}}(|1\rangle_A|0\rangle_B + e^{i\phi_{AB}}|0\rangle_A|1\rangle_B)$$

Heralded Entangled number state of remote QM

Measurement-Induced Entanglement for Excitation Stored in Remote Atomic Ensembles,

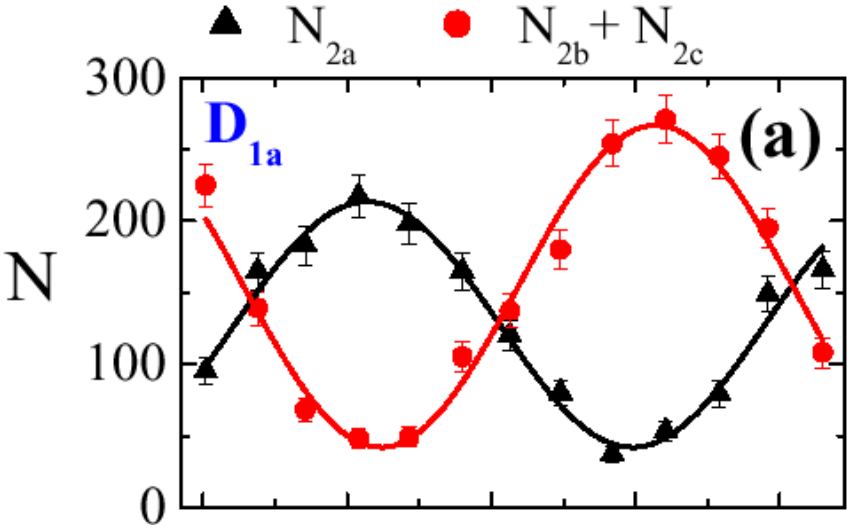
C.W. Chou, H. de Riedmatten, D. Felinto, S.V. Polyakov, S.van Enk, & H.J. Kimble, Nature 438, 828 (2005)



Conditions for entanglement :

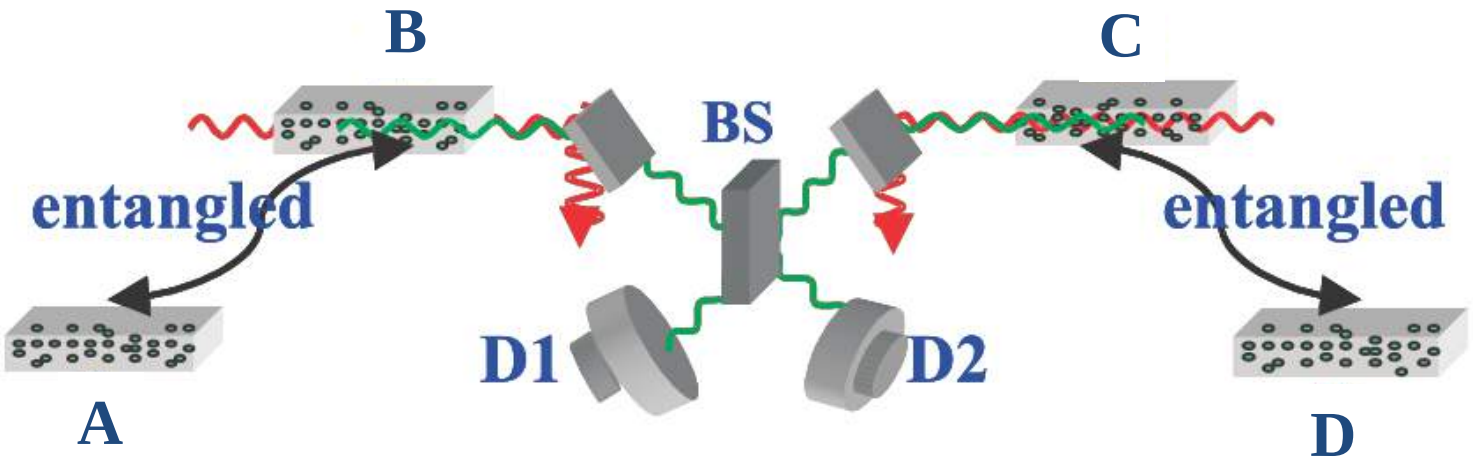
-single excitation regime (p<<1)

-coherent superposition



1 quantum of excitation
shared in an entangled quantum state
between two atomic ensembles
located ~ 3 meters apart

Entanglement connection



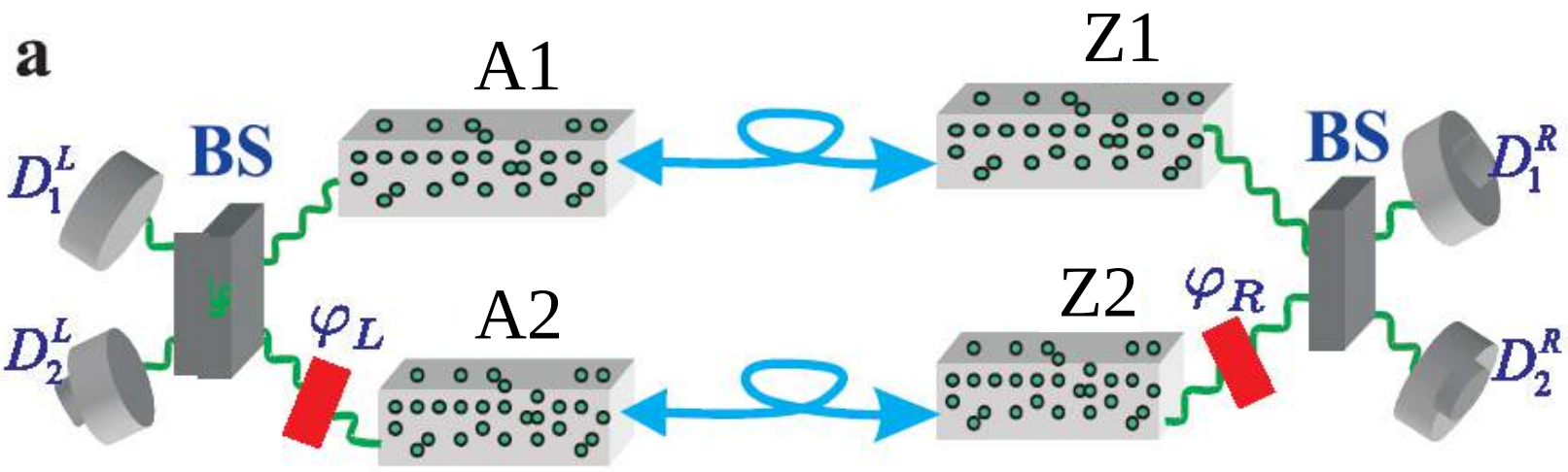
$$|\Phi\rangle = \frac{1}{2}\left(|1\rangle_A|0\rangle_B + e^{i\theta_{AB}}|0\rangle_A|1\rangle_B\right)\otimes\left(|1\rangle_C|0\rangle_D + e^{i\theta_{CD}}|0\rangle_C|1\rangle_D\right)$$

$$|\psi_{ad}\rangle = \frac{1}{\sqrt{2}}(s_a^\dagger + s_d^\dagger e^{i(\theta_{ab}+\theta_{cd})})|0\rangle$$

$$\rho_{ad} = \alpha_1|\psi_{ad}\rangle\langle\psi_{ad}| + \beta_1|0\rangle\langle 0|$$

DLCZ quantum repeater

- Number state entanglement not practical for quantum information manipulation (need for phase reference, difficult to manipulate)
- DLCZ Solution: Implement two chains of entangled ensembles. Effective polarization like entangled state



$$\left(a_1^+ + e^{i\theta_1} z_1^+\right) \otimes \left(a_2^+ + e^{i\theta_2} z_2^+\right) \left|0\right\rangle$$

$$= \left(a_1^+ a_2^+ + e^{i\theta_1} z_1^+ a_2^+ + e^{i\theta_2} z_2^+ a_1^+ + e^{i\theta_1} z_1^+ e^{i\theta_2} z_2^+\right) \left|0\right\rangle$$

Post-selected state

$$\left(z_1^+ a_2^+ + e^{i(\theta_2 - \theta_1)} z_2^+ a_1^+\right) \left|0\right\rangle$$

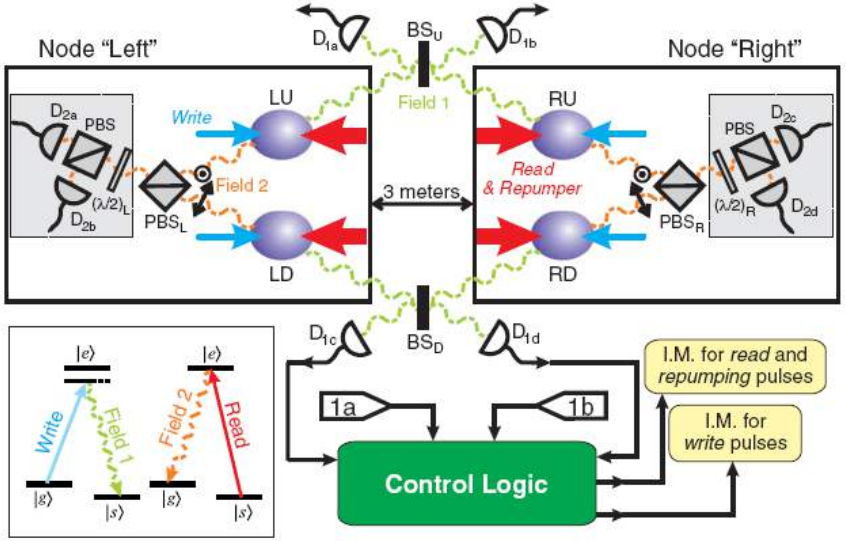
L.-M. Duan, M.D. Lukin, J.I. Cirac, and P. Zoller, Nature 414, 413 (2001).

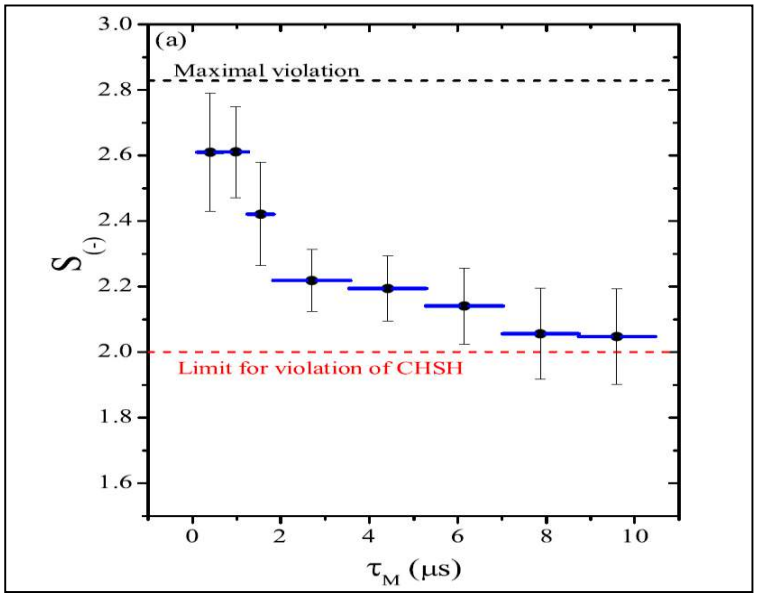
Elementary Segments of Quantum Repeaters with Atomic ensembles

Functional Quantum Nodes for Entanglement Distribution over Scalable Quantum Networks

SCIENCE VOL 316 1 JUNE 2007

Chin-Wen Chou, Julien Laurat, Hui Deng, Kyung Soo Choi, Hugues de Riedmatten,*
Daniel Felinto,† H. Jeff Kimble‡





Experimental demonstration of a BDCZ quantum repeater node

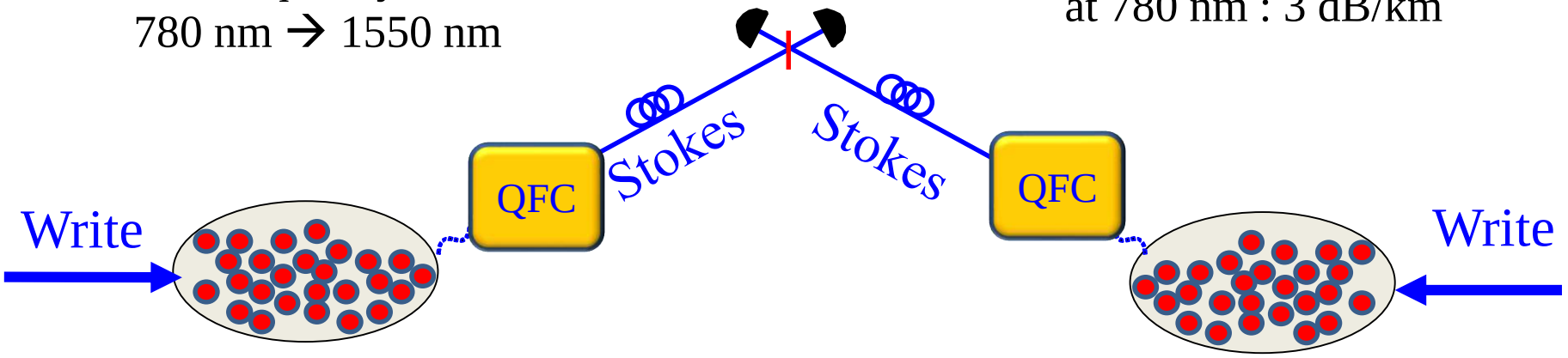
Zhen-Sheng Yuan[1,2]*, Yu-Ao Chen[1,2]*, Bo Zhao[1], Shuai Chen[1], Jörg Schmiedmayer[3] & Jian-Wei Pan[1,2]

Nature 454, 1098 (2008)

Entanglement between remote atomic ensembles

Quantum Frequency conversion
780 nm → 1550 nm

Absorption in fibers
at 780 nm : 3 dB/km



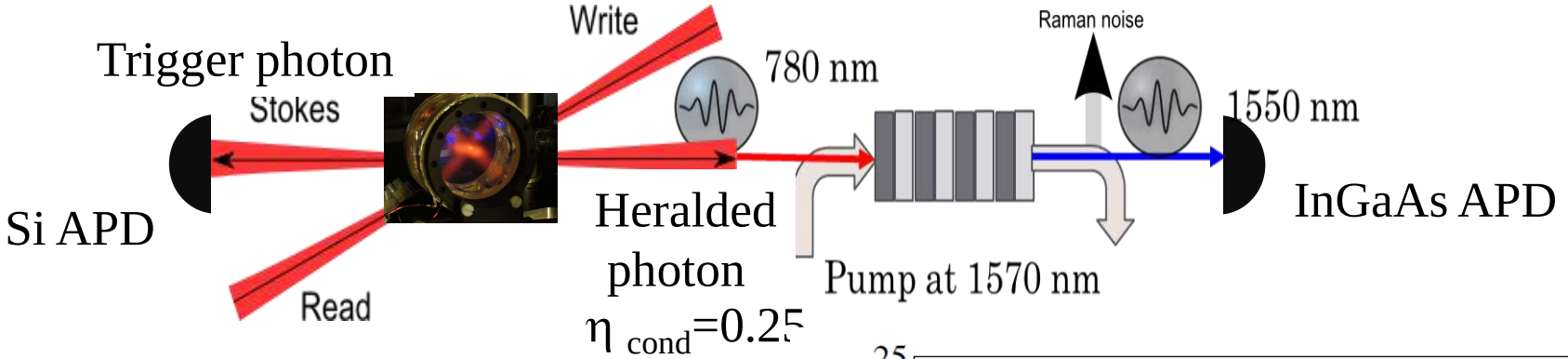
- Four Wave mixing in ultra dense cold atoms

  (Radnaev *et al*, Nature Physics (2010))
  Entanglement between telecom photon (1367nm) and long lived spin-wave (10 ms)

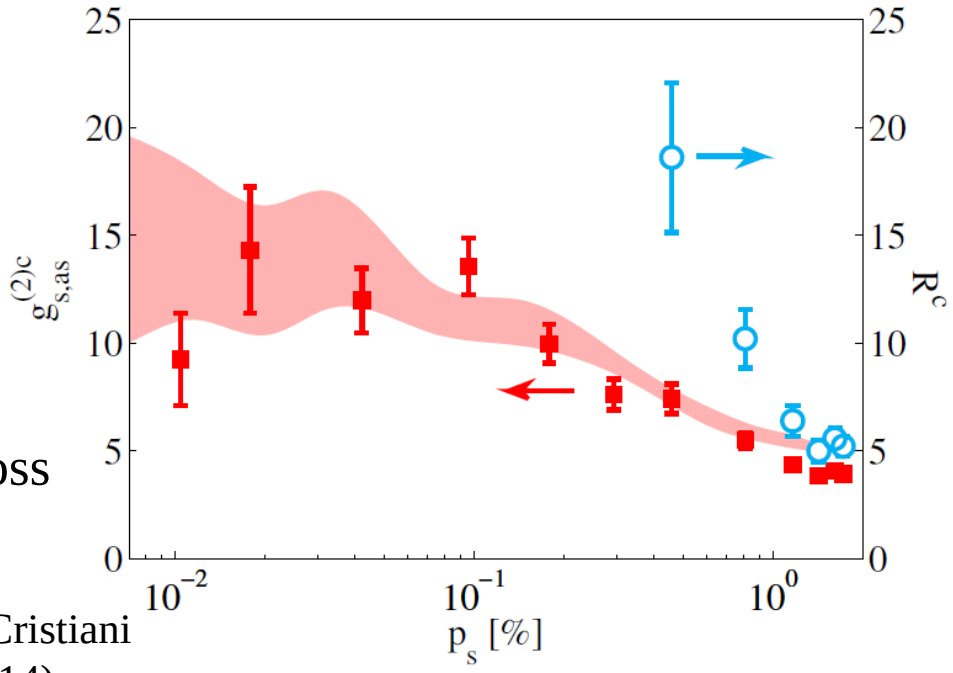
- Our approach : Integrated Photonic Interface using Non –Linear Waveguide

Combining DLCZ QM and QFC

- DLCZ quantum memory used as an heralded single photon source
- Single photons converted through QFC setup

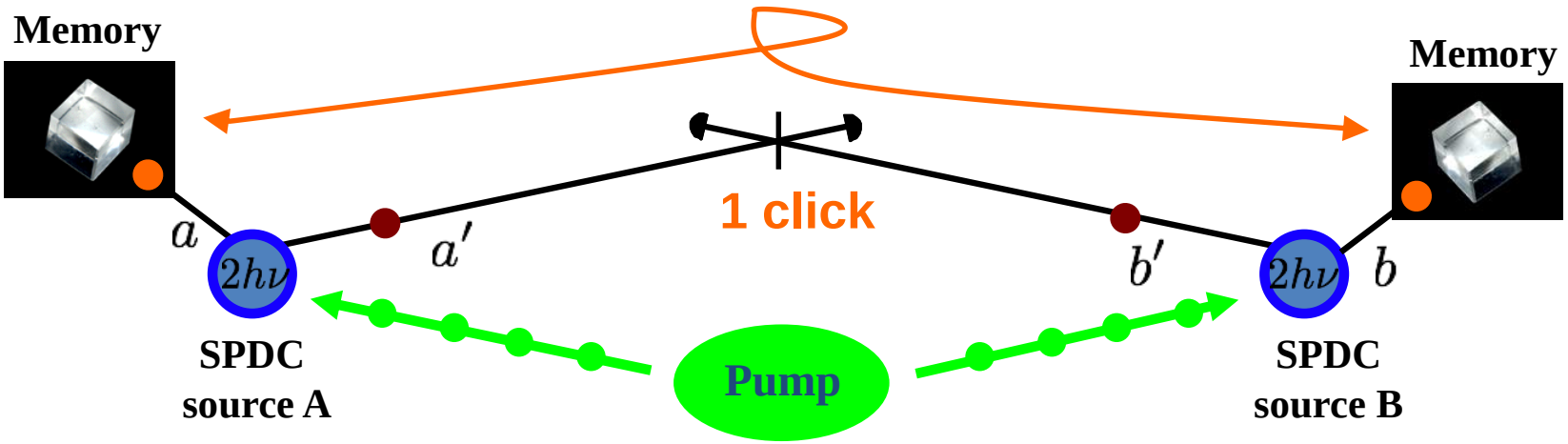


- Highly non-classical correlations after conversion
  (Only degraded due to pump noise)
- Internal conversion efficiency of single photons :76 %
  Device efficiency limited by optical loss



B. Albrecht, P. Farrera, X. Fernandez-Gonzalvo, M. Cristiani and H. de Riedmatten , Nature Commun. 5, 3376 (2014)

Photon-pair sources and solid-state QMs



$$\left(1+\sqrt{\frac{p}{2}}(a^\dagger a'^\dagger + b^\dagger b'^\dagger)+O(p)\right)|0\rangle$$

Initial state

$$|\Phi_{AB}\rangle = \frac{1}{\sqrt{2}}(a^\dagger e^{i\phi_A} + b^\dagger e^{i\phi_B})|0\rangle$$

$$|\Phi_{AB}\rangle = \frac{1}{\sqrt{2}}(|1\rangle_A|0\rangle_B + e^{i\phi_{AB}}|0\rangle_A|1\rangle_B)$$

$$\phi_{AB} = \phi_B - \phi_A$$

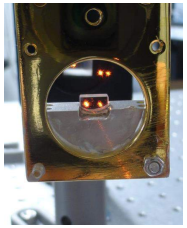
Conditional state (one click!)

Heralded entangled state of remote QM

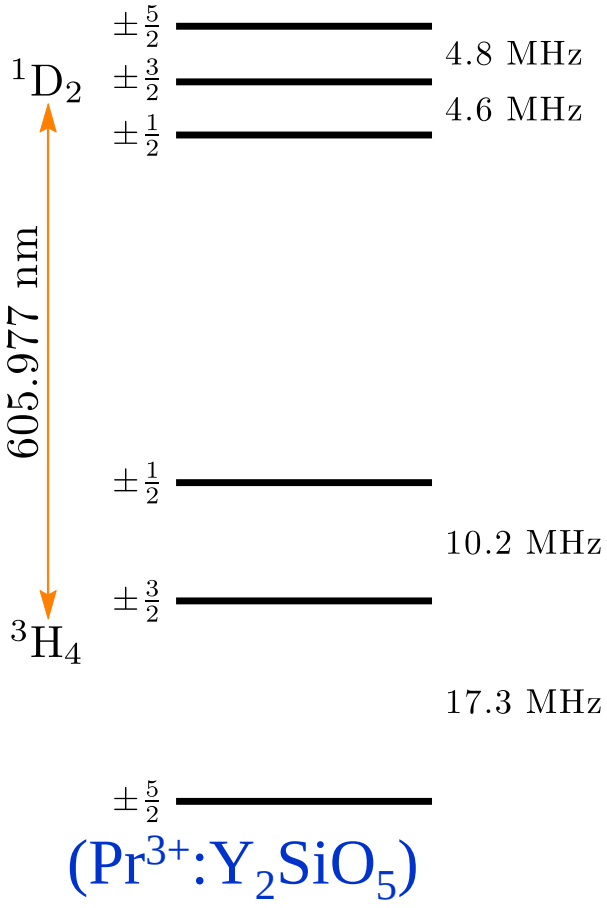
C. Simon, H. de Riedmatten, M.Afzelius, N. Sangouard, H. Zbinden and N. Gisin, PRL **98**, 190503 (2007)

Rare-earth ion doped crystals

- Large number of stationary atoms with optical and spin transitions.
- Excellent coherent properties (T<4K)
- Static inhomogeneous broadening (~ GHz) which can be tailored.



Candidate for QMs : Nd, Er, Eu, Pr , Tm



Optical T_2	111µs
Spin T_2	> 1 s

Light storage (with bright pulses):
on the order of seconds
J.J. Longdell et. al., PRL **95**, 063601 (2005)
on the order of minutes
G. Heinze et. al., PRL **111**, 033601 (2013)
Quantum storage of weak coherent states with 69% storage and retrieval efficiency
M.P. Hedges, et. al., Nature, **465** 1052, (2010)

Advantages:

-Long spin coherence time

-High absorption (α=20 cm^{-1})

-Good level structure for spin state storage

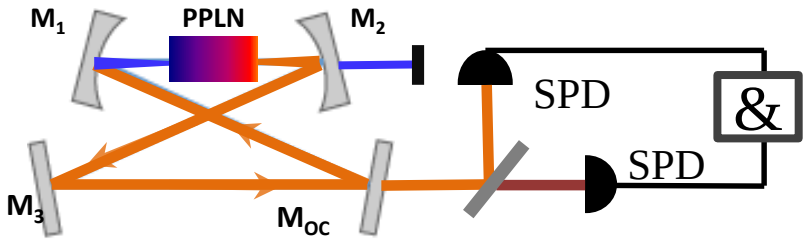
Drawbacks:

-small bandwidth (<4 MHz) : quantum light source challenging

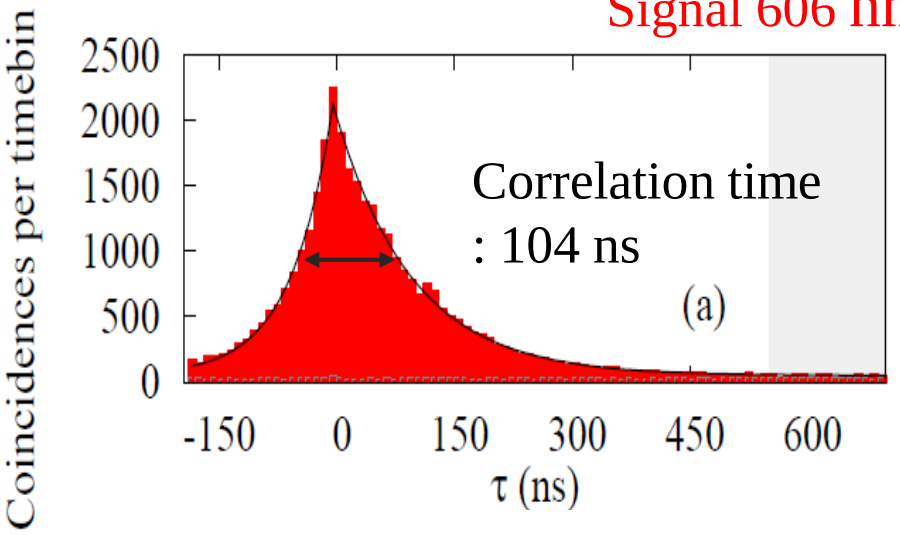
-wavelength (606nm)

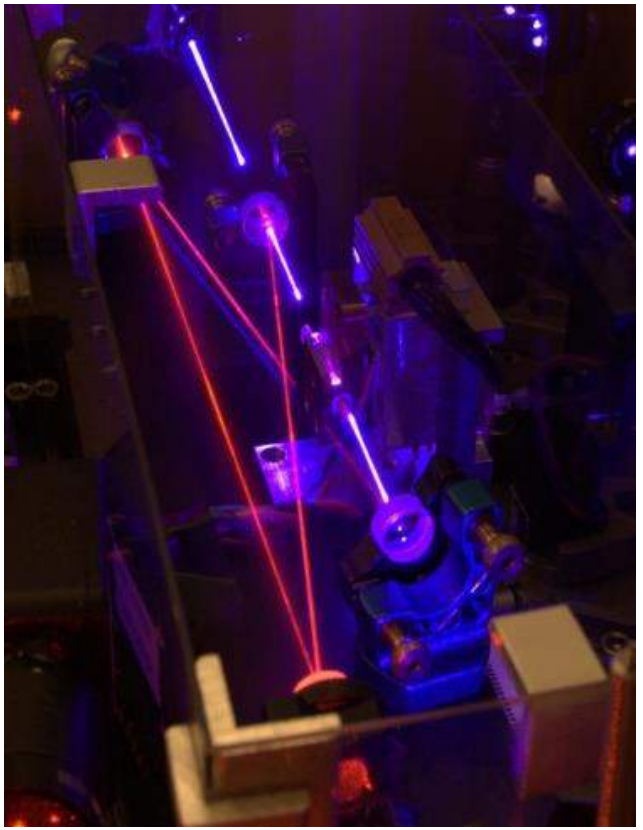
Ultra-Narrowband Photon Pair source compatible with Solid-State QM and Telecom fibers

Idler 1436 nm



Signal 606 nm



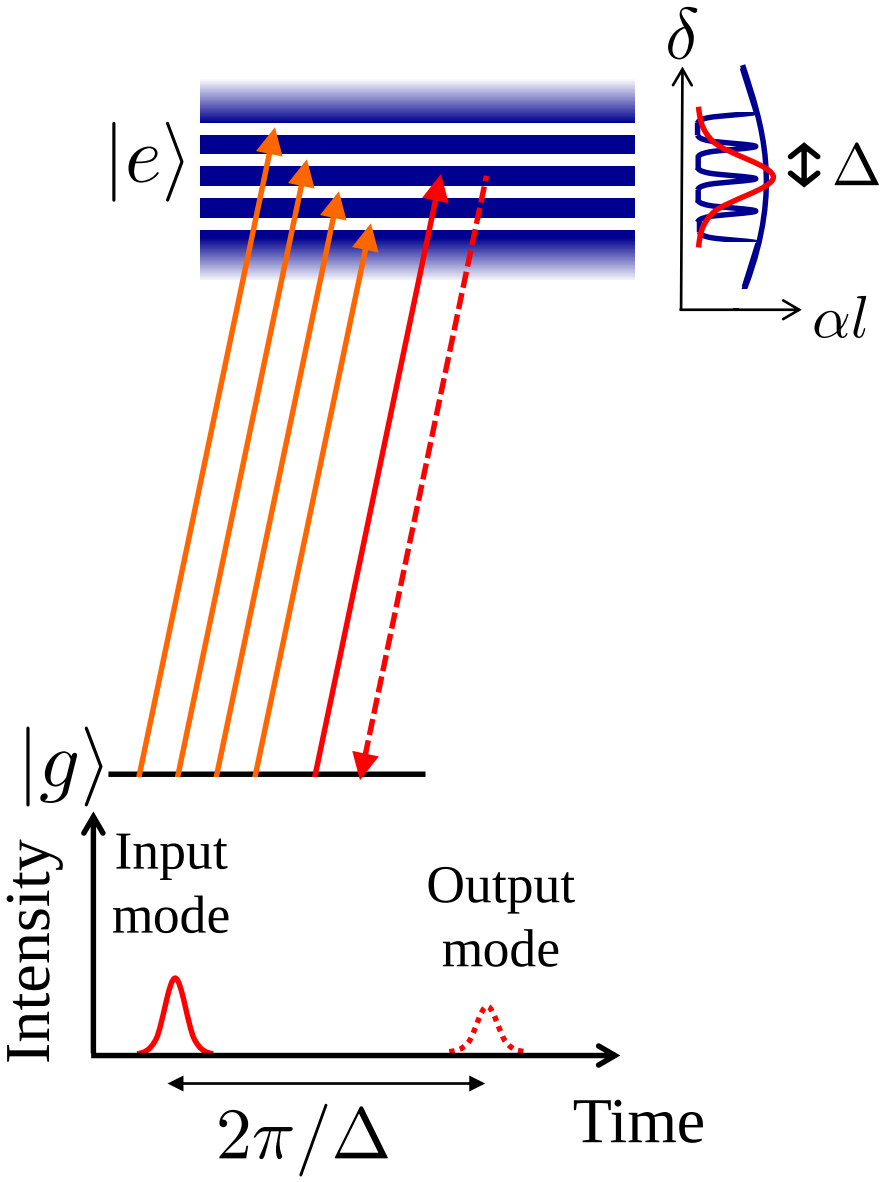


Cavity enhanced SPDC
Photon pairs with linewidth 2-3 MHz

See also other work at Geneva, Calgary, Erlangen

J. Fekete, D. Rieländer, M. Cristiani, H. de Riedmatten, Phys. Rev. Lett. **110**, 220502 (2013)

Atomic Frequency Comb (AFC) Memory



State after mapping

$$\sum_{k=1}^{N} c_k \left| g_1 g_2 ... e_k ... g_N \right\rangle$$

Dephasing

$$\sum_{k=1}^{N} c_k e^{-i\delta_k t} \left| g_1 g_2 ... e_k ... g_N \right\rangle$$

$$\delta_k = m_k \Delta \rightarrow e^{-i(m_k \Delta)t}$$

Rephasing after a time

$$t_e = \frac{2\pi}{\Delta}$$

Collective, coherent emission in the forward mode. Photon echo-like emission.

Noise free.

M. Afzelius, C. Simon, H. de Riedmatten, and N. Gisin, Phys.Rev.A 79, 052329 (2009)

Entanglement between a photon and a solid state quantum memory

LETTER

Nature **469**, 508 (2011)

doi:10.1038/nature09662

Quantum storage of photonic entanglement in a crystal

$Nd^{3+}:Y_2SiO_5$

Christoph Clausen[1]*, Imam Usmani[1]*, Félix Bussières[1], Nicolas Sangouard[1], Mikael Afzelius[1], Hugues de Riedmatten[1,2,3] & Nicolas Gisin[1]

Nature **469**, 512 (2011)

LETTER

doi:10.1038/nature09719

Broadband waveguide quantum memory for entangled photons

$Tm^{3+}:LiNbO_3$

Erhan Saglamyurek[1], Neil Sinclair[1], Jeongwan Jin[1], Joshua A. Slater[1], Daniel Oblak[1], Félix Bussières[1]†, Mathew George[2], Raimund Ricken[2], Wolfgang Sohler[2] & Wolfgang Tittel[1]

Entanglement between a photon and a solid state quantum memory

LETTER

Nature **469**, 508 (2011)

doi:10.1038/nature09662

Quantum storage of photonic entanglement in a crystal

$Nd^{3+}:Y_2SiO_5$

Christoph Clausen[1]*, & Nicolas Gisin[1]

Quantum light storage only in excited state (Ts max =200 ns) and in two level ground state systems :
not compatible with AFC spin wave storage

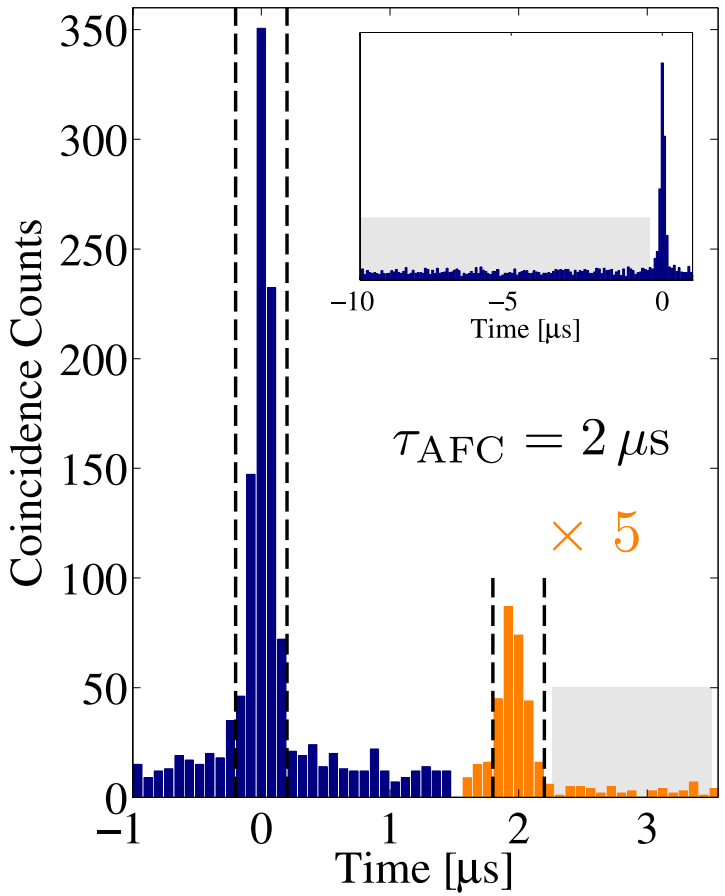
LETTE

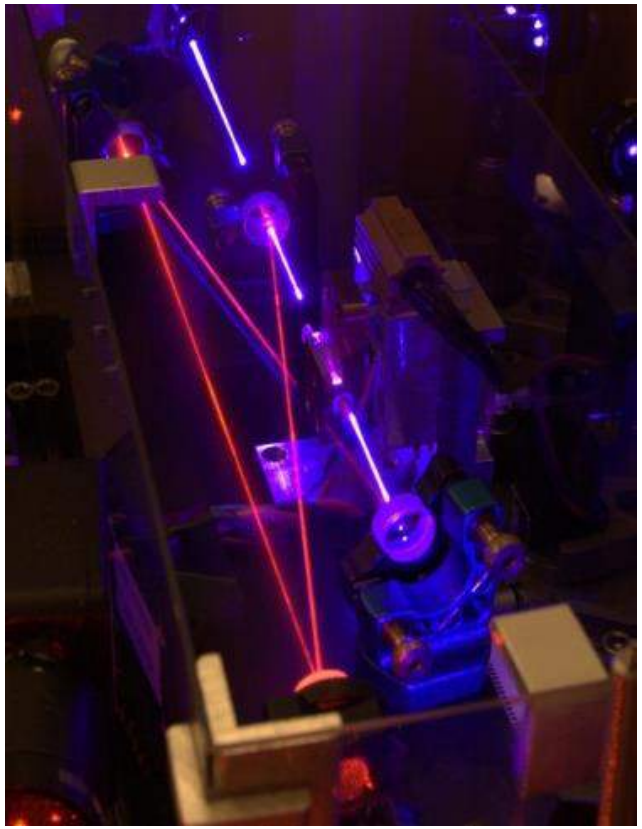
Broadband waveguide quantum memory for entangled photons

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Storing Quantum Light in a $Pr^{3+}:Y_2SiO_5$ Memory





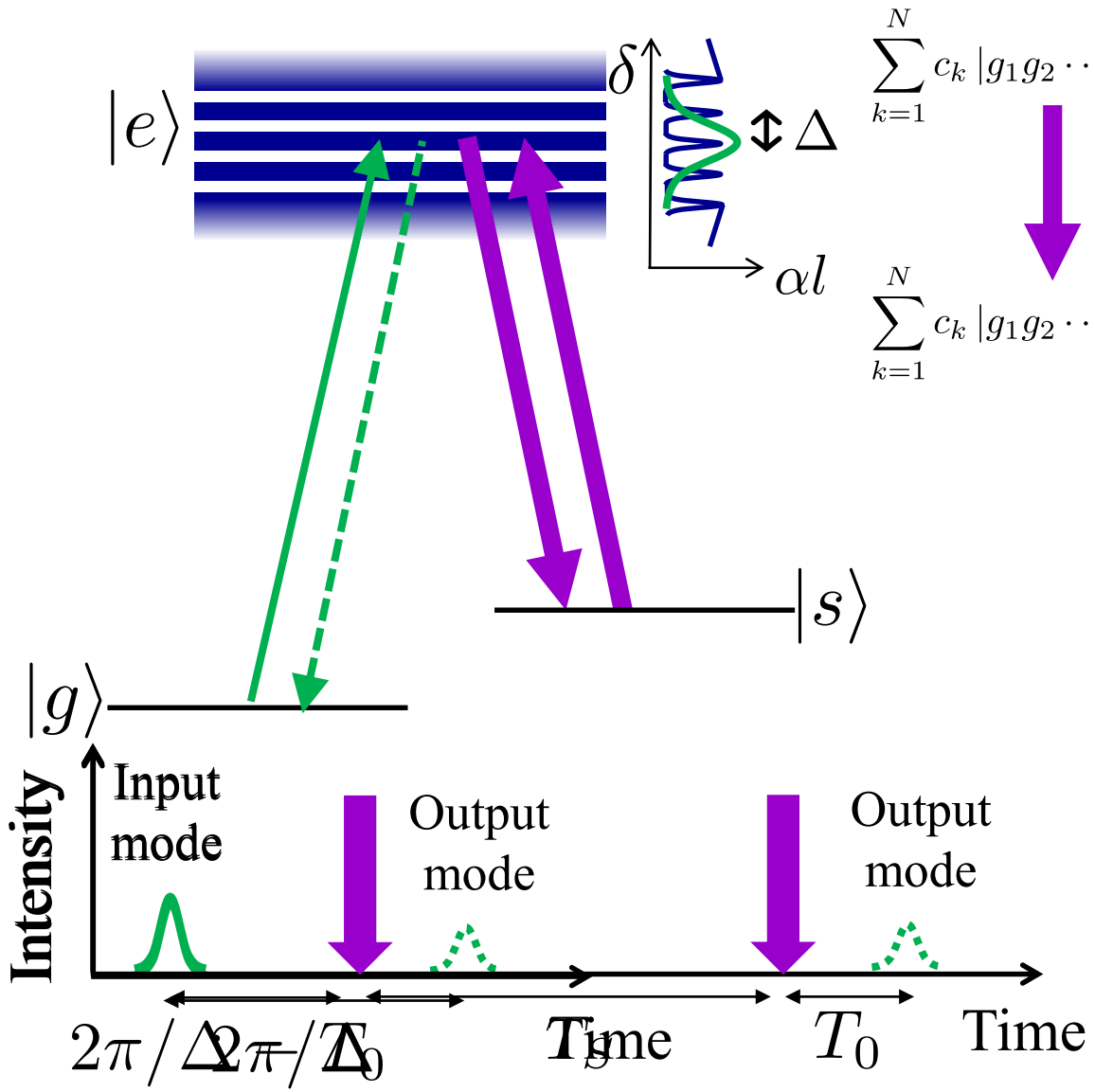
Heralded single photon storage
5-10 % efficiency, up to 5 us

Cavity enhanced SPDC
Photon pairs with linewidth 2-3 MHz
1 photon at telecom 1 QM resonant

J. Fekete, D. Rieländer, M. Cristiani, H. de Riedmatten, Phys. Rev. Lett. **110**, 220502 (2013)

D. Rieländer, K.Kutluer, P.M. Ledingham, M. Gundogan, J. Fekete, M. Mazzera, H. de Riedmatten, PRL. **112**, 040504 (2014)

Spin-Wave AFC Protocol



$$\sum_{k=1}^{N} c_k \, |g_1 g_2 \cdots e_k \cdots g_N\rangle$$

$$\sum_{k=1}^{N} c_k \, |g_1 g_2 \cdots s_k \cdots g_N\rangle$$

- On-demand read-out
- Longer storage times

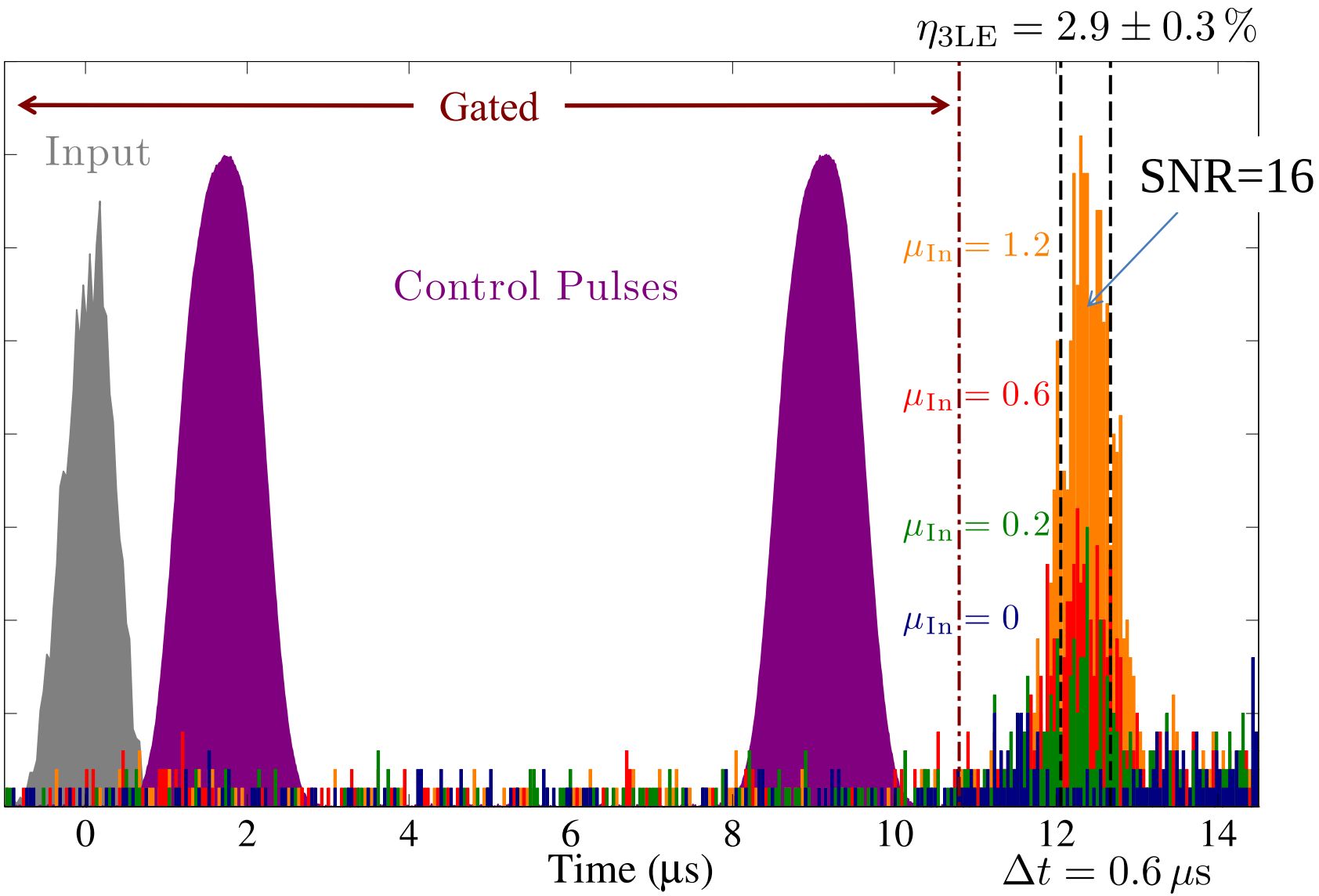
Bright pulses

M. Afzelius *et al*, PRL, 104, 040503(2010).

Single photon level :

Challenge due technical noise from the control pulses

AFC spin-wave storage at single photon level

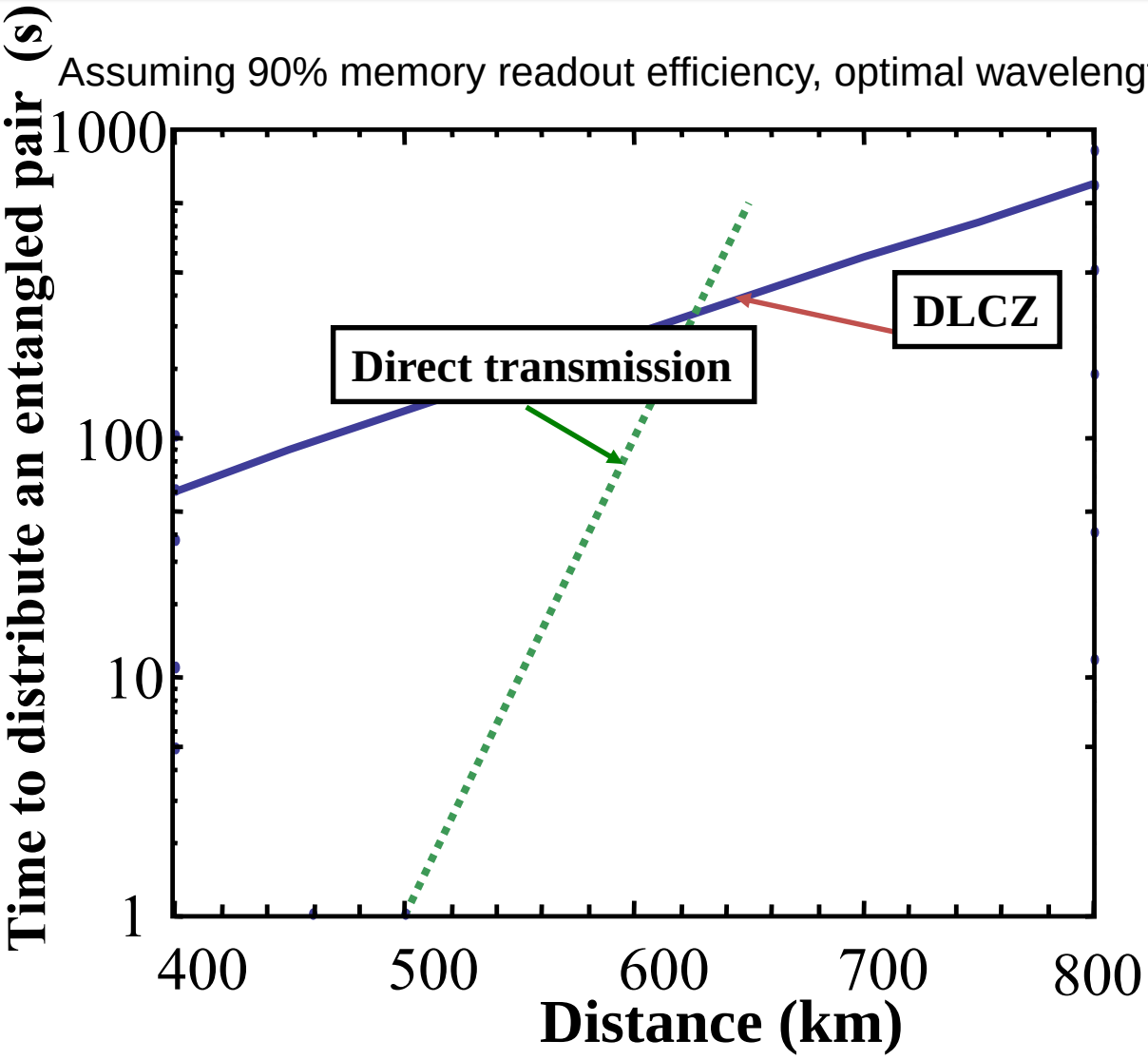


M. Gündoğan, P. M. Ledingham, K. Kutluer, M. Mazzera, H. de Riedmatten, PRL 114, 230501 (2015)

See also:P. Jobez, C. Laplane, N. Timoney, N. Gisin, A. Ferrier, P. Goldner, M. Afzelius,PRL 114, 230502 (2015

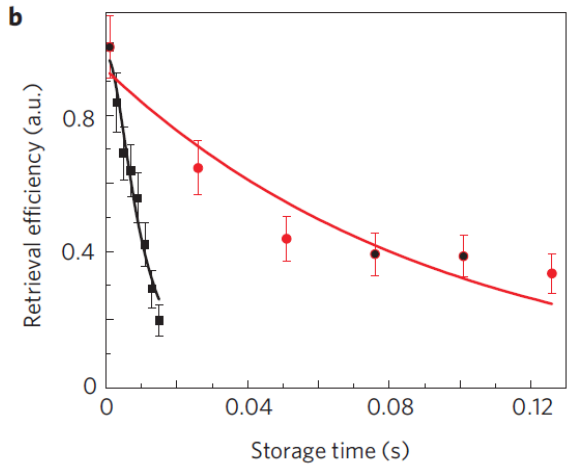
Q repeaters with atomic ensembles and linear optics

Assuming 90% memory readout efficiency, optimal wavelength (1550 nm).

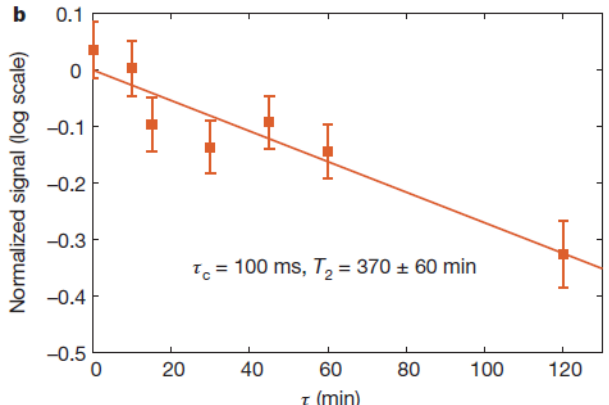


N. Sangouard, C.Simon, H. de Riedmatten and N.Gisin, RMP (2011)

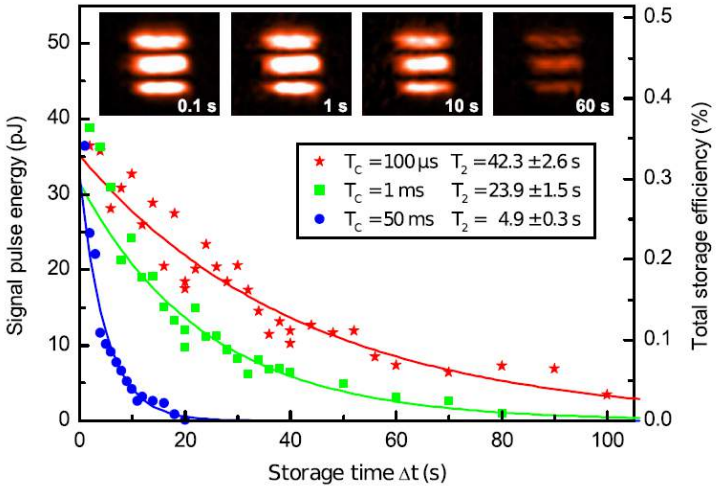
Long-lived quantum memories ?



EIT coherent optical memory:
Storage time 1 minute
Pr:YSO crystal
Heinze *et al*, PRL 2013



DLCZ quantum memory:
Storage time 100 ms
Cold Rb atoms in optical lattice
Kuzmich Nature Phys 2010



Spin coherence : 6 hours (no light storage yet)
Eu:YSO crystal
Sellars, Nature 2015

Limitations of repeaters based on atomic ensembles

- Probabilistic light-matter entanglement
- Communication time (due to heralded entanglement)
- Probabilistic Bell state measurements

Problem : DLCZ or SPDC : probabilistic sources

$$\left|\Phi_{A,S}\right\rangle = \left|0_A,0_S\right\rangle + \sqrt{p}\left|1_A,1_S\right\rangle + p\left|2_A,2_S\right\rangle + O(p^{3/2})$$

Cross-correlation function

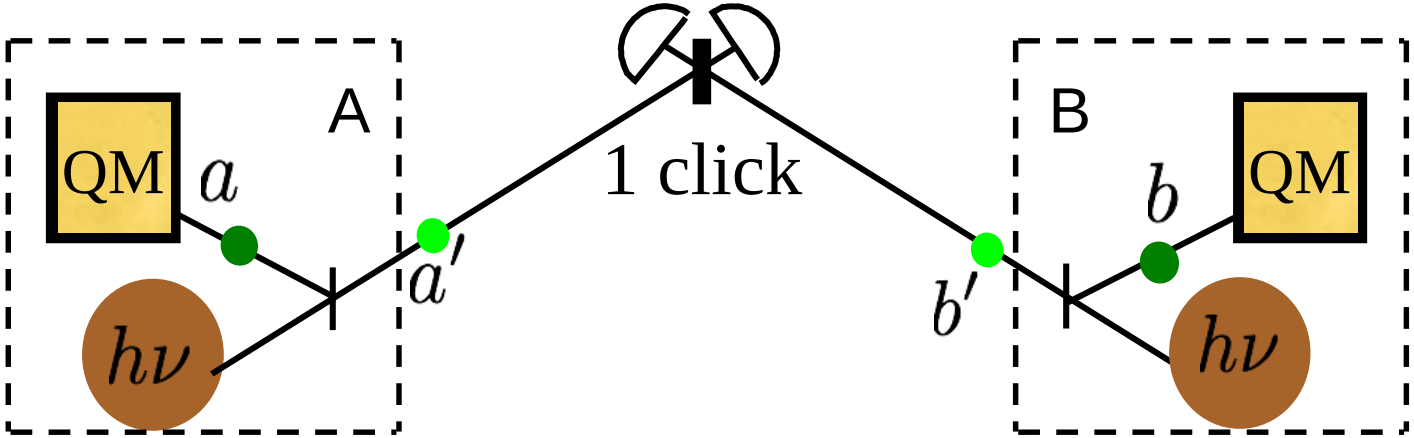
$$g^{(2)}_{S,AS}(\tau) = \frac{\left\langle \hat{I}_S(t)\hat{I}_{AS}(t+\tau)\right\rangle}{\left\langle \hat{I}_S\right\rangle\left\langle \hat{I}_{AS}\right\rangle} = \frac{p_{S,AS}}{p_S\, p_{AS}} = 1+\frac{1}{p}$$

Entanglement fidelity :

$$F = \frac{g^{(2)}_{S,AS}(\tau)}{g^{(2)}_{S,AS}(\tau)+1}$$

Trade-off between fidelity and efficiency

Q repeaters with single photon sources and quantum memories



Initial state $(\alpha a^\dagger + \beta a'^\dagger)(\alpha b^\dagger + \beta b'^\dagger)|0\rangle$

$(\alpha^2 a^\dagger b^\dagger + \alpha\beta(a^\dagger b'^\dagger + a'^\dagger b^\dagger) + \beta^2 a'^\dagger b'^\dagger)|0\rangle$

Conditional state $\beta^2|0\rangle\langle 0| + \alpha^2|\psi_+\rangle\langle\psi_+|$

with $|\psi_+\rangle = \frac{1}{\sqrt{2}}(a^\dagger + b^\dagger)|0\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$

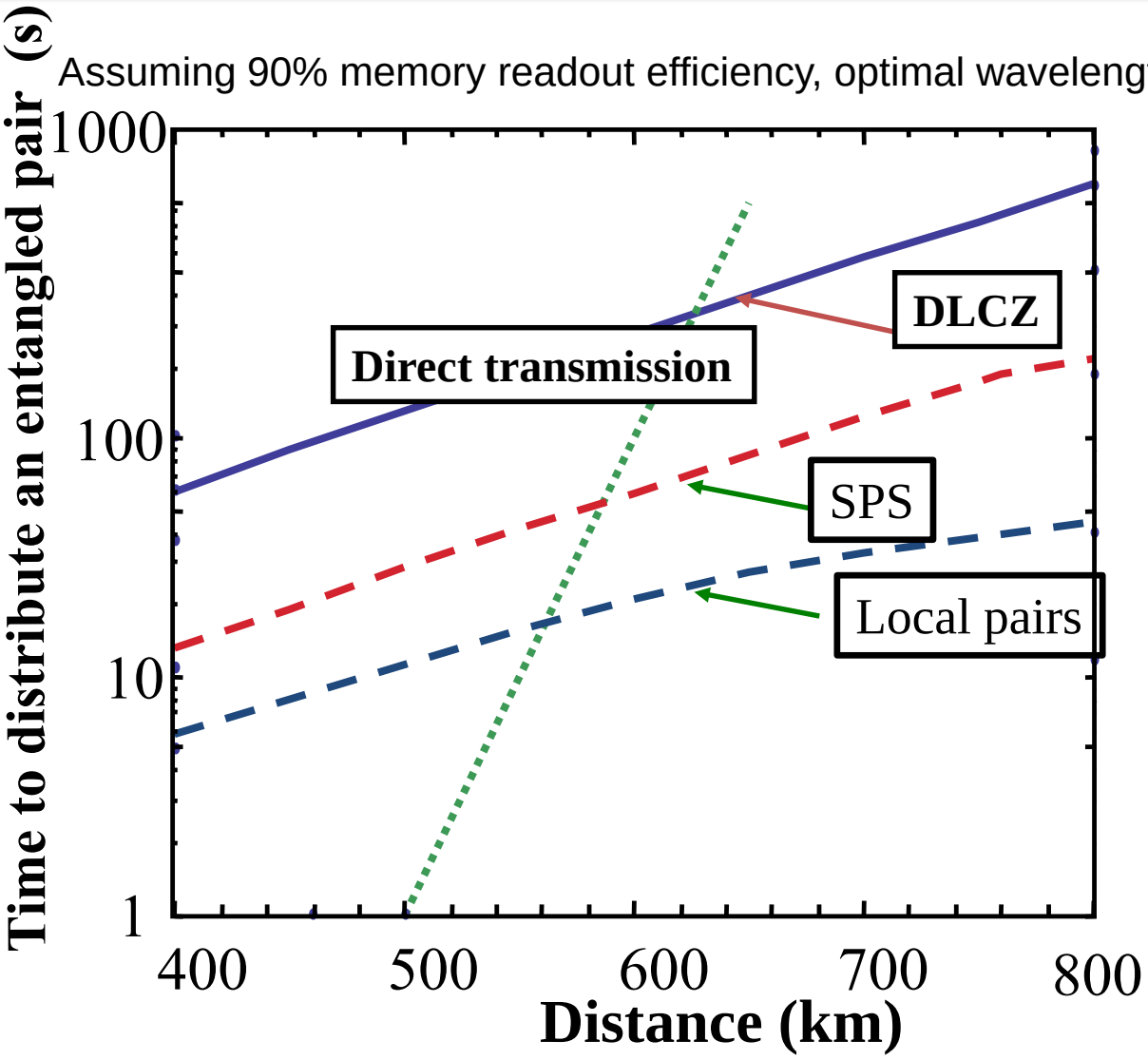
Entangled state of two memories

Detection of empty modes with post-selection

N. Sangouard, C. Simon, J.Minář, H. Zbinden, H. de Riedmatten, and N. Gisin, PRA 76, 050301 (R) (2007)

Q repeaters with atomic ensembles and linear optics

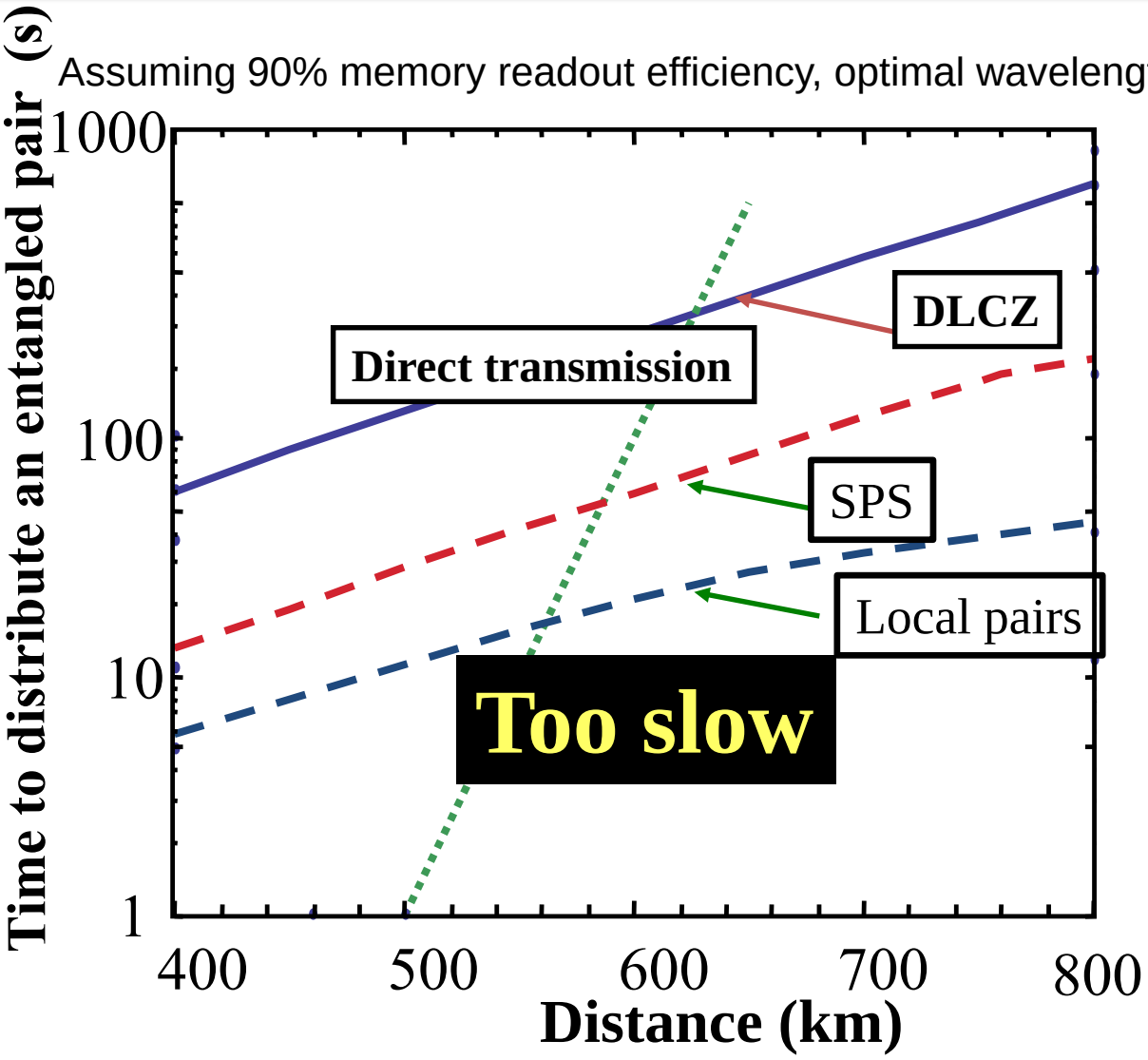
Assuming 90% memory readout efficiency, optimal wavelength (1550 nm).



N. Sangouard, C.Simon, H. de Riedmatten and N.Gisin, RMP (2011)

Q repeaters with atomic ensembles and linear optics

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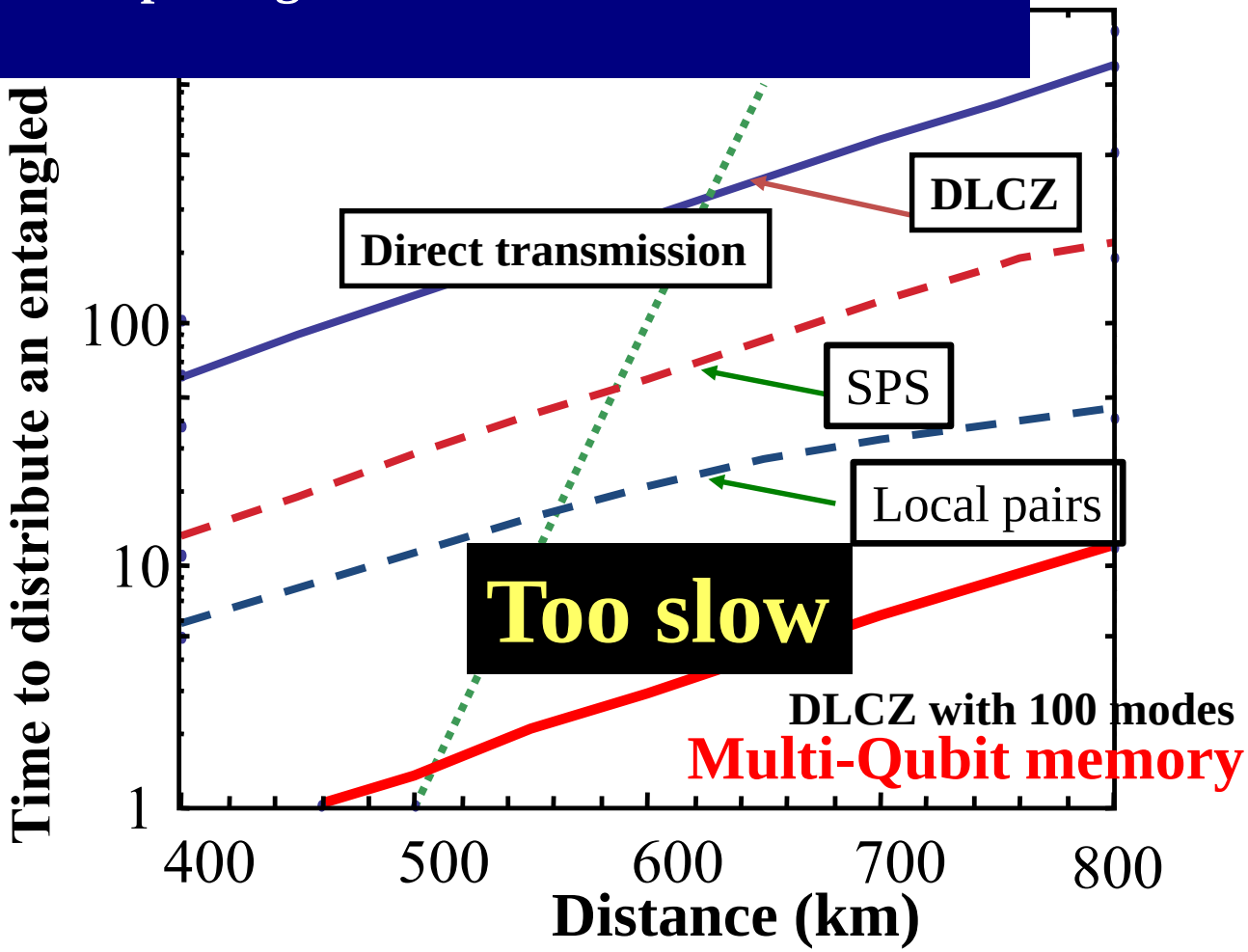


N. Sangouard, C.Simon, H. de Riedmatten and N.Gisin, RMP (2011)

Q repeaters with atomic ensembles and linear optics

Storing N modes in <u>ONE</u> memory using *<u>time</u>*, spatial or frequency multiplexing will reduce this time with a factor N !

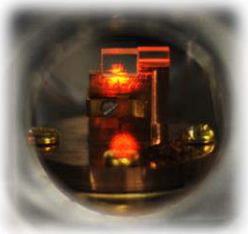
avelength (1550 nm).



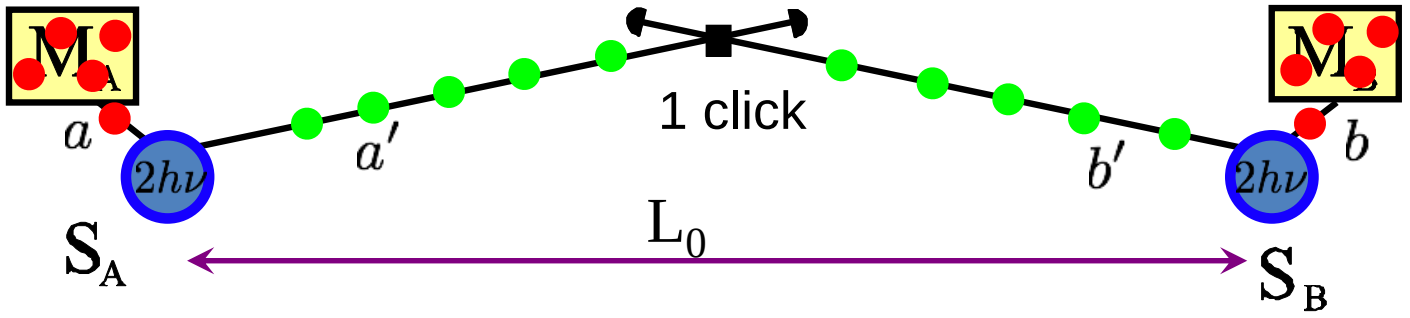
N. Sangouard, C.Simon, H. de Riedmatten and N.Gisin, RMP (2011)

Outline

- Introduction to quantum repeaters
- Quantum repeaters based on atomic ensembles and linear optics
  - The DLCZ protocol
  - Coupling cold atomic QMs to telecom wavelengths
  - Photon pair source and solid-state quantum memories
  - More deterministic schemes
- Multiplexed quantum repeaters
- Quantum repeaters based on single emitters



Entanglement generation with multimode memories



Conventional memory: have to wait time L_0/c before trying again.
(Ex. For 100 km, L_0/c=500 us, R=2 kHz)

$$P_0 = p\eta_{L_0}\eta_D$$ Low success probability! (Typ. 10^{-3} - 10^{-4})

Memories that can store *N* temporal modes.

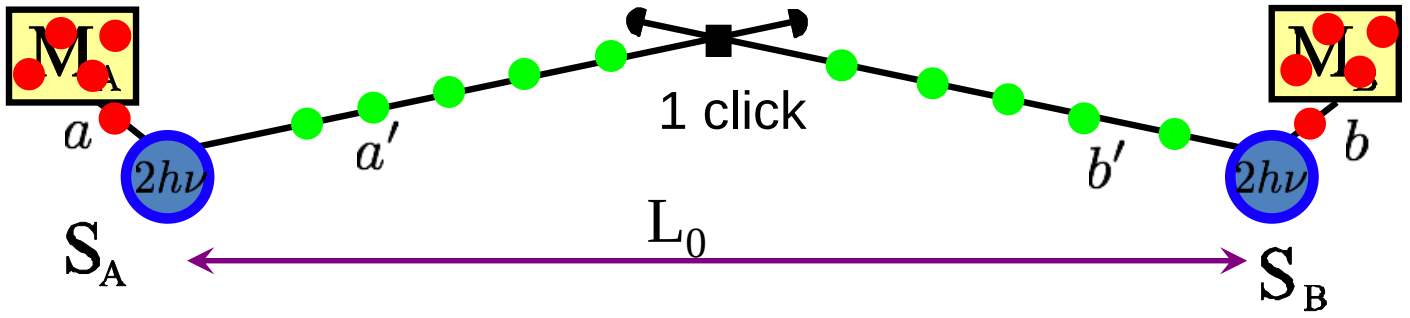
N attempts per time interval L_0/c

$$P_0^{(N)} = 1-(1-P_0^{(1)})^N \approx NP_0^{(1)} \qquad (N > 100 \text{ possible})$$

Speedup by factor of *N*.

C. Simon, H. de Riedmatten, M. Afzelius, N. Sangouard, H. Zbinden and N. Gisin
Phys. Rev. Lett. 98, 190503 (2007)

Entanglement generation with multimode memories



Conventional memory: have to wait time L_0/c before trying again.
(Ex. For 100 km, L_0/c=500 us, R=2 kHz)

The memories need to be able to

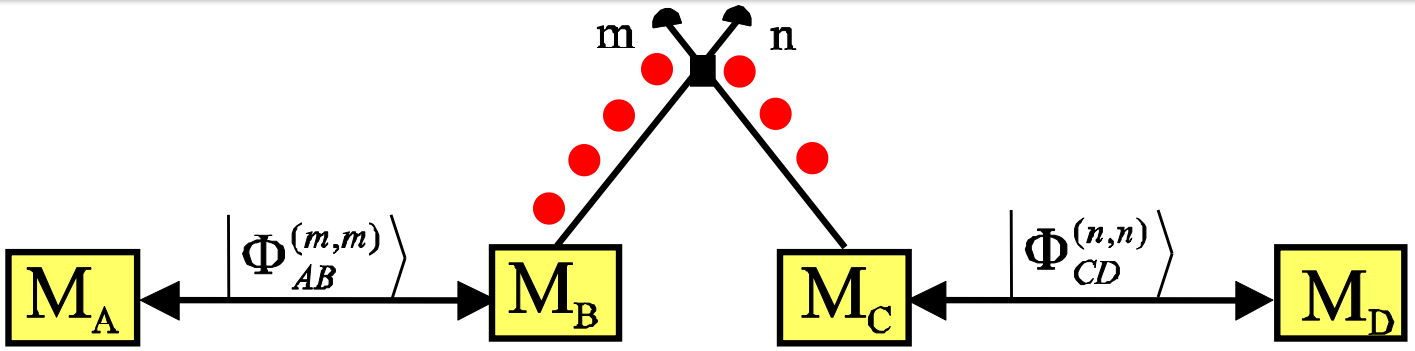
- store N temporally distinguishable modes
- selective read-out
- preserve the phase of each mode

$P_0' = 1-(1-P_0')^N \approx N P_0'$ *(N > 100 possible)*

Speedup by factor of *N*.

C. Simon, H. de Riedmatten, M. Afzelius, N. Sangouard, H. Zbinden and N. Gisin
Phys. Rev. Lett. 98, 190503 (2007)

Entanglement connection with multi-mode memories



Entanglement stored in different memory modes for the two links.

$$|\Phi_{AB}^{(m,m)}\rangle = \frac{1}{\sqrt{2}}(a_m^\dagger + b_m^\dagger e^{i\phi_{AB}})|0\rangle$$

$$|\Phi_{CD}^{(n,n)}\rangle = \frac{1}{\sqrt{2}}(a_n^\dagger + b_n^\dagger e^{i\phi_{CD}})|0\rangle$$

Initial states

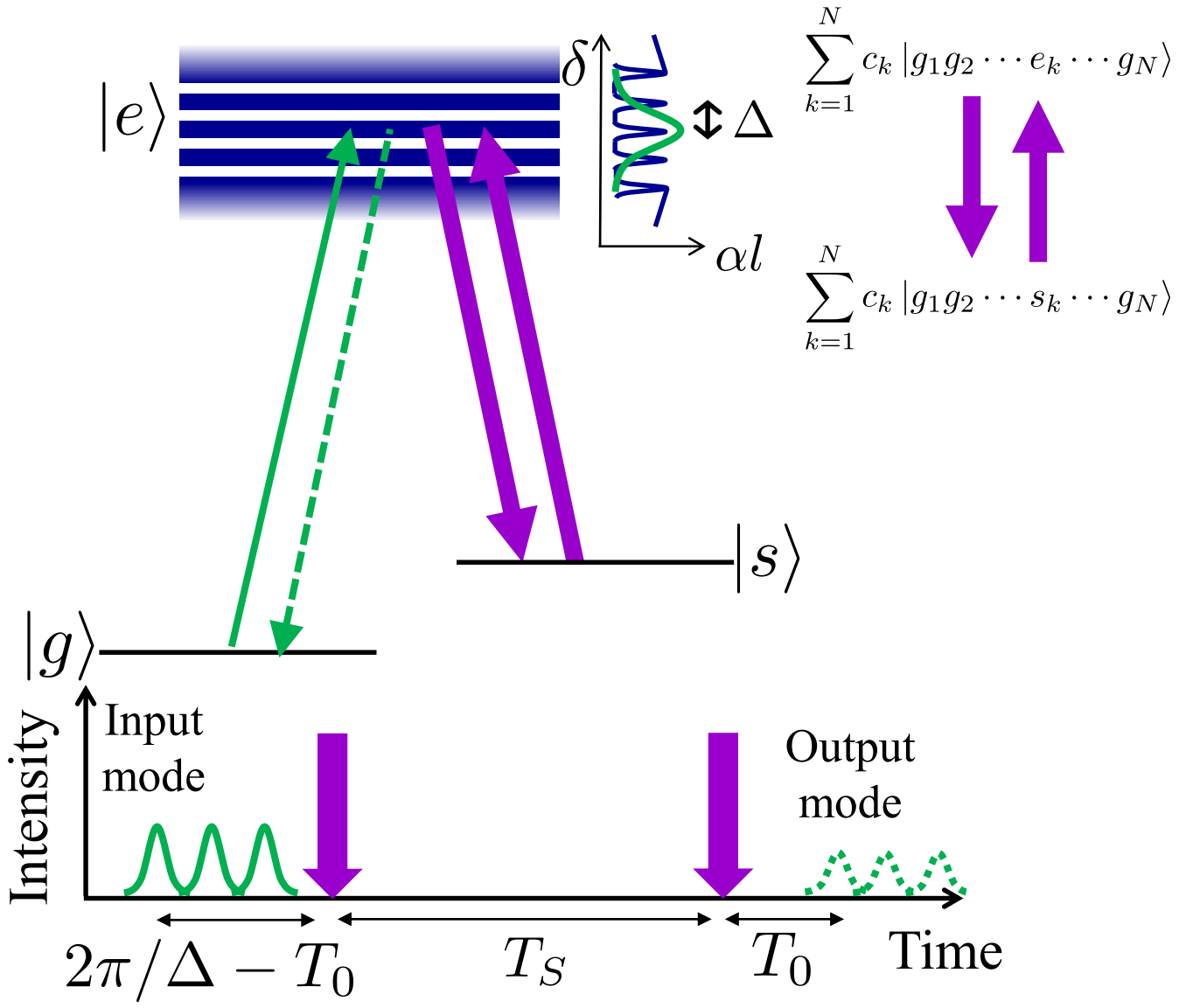
$$|\Phi_{AD}^{(m,n)}\rangle = \frac{1}{\sqrt{2}}(a_m^\dagger + d_n^\dagger e^{i(\phi_{AB}+\phi_{CD})})|0\rangle$$

Conditional state

Need to store and retrieve *N* modes, preserving their distinguishability.

C. Simon, H. de Riedmatten, M. Afzelius, N. Sangouard, H. Zbinden and N. Gisin
Phys. Rev. Lett. 98, 190503 (2007)

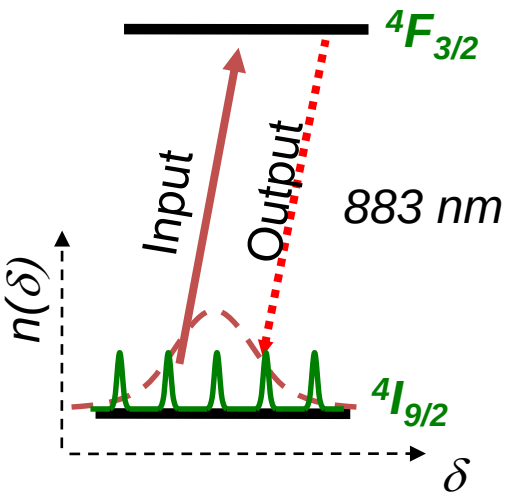
AFC Protocol: a temporal multimode memory

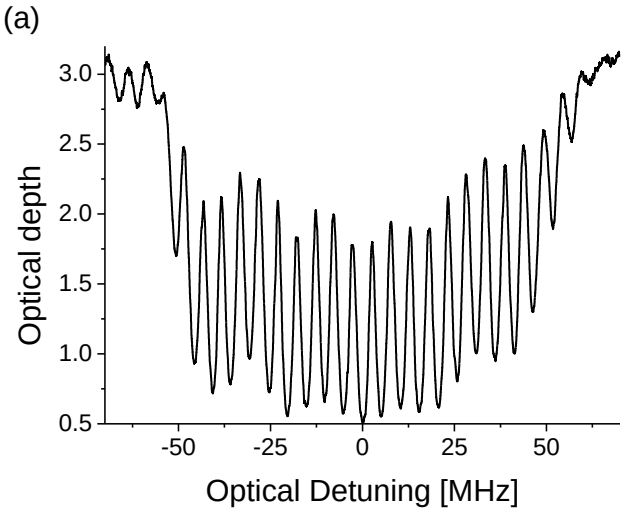


Multi-mode storage in Nd3+:Y2SiO5

⟨n⟩ < 1 per mode

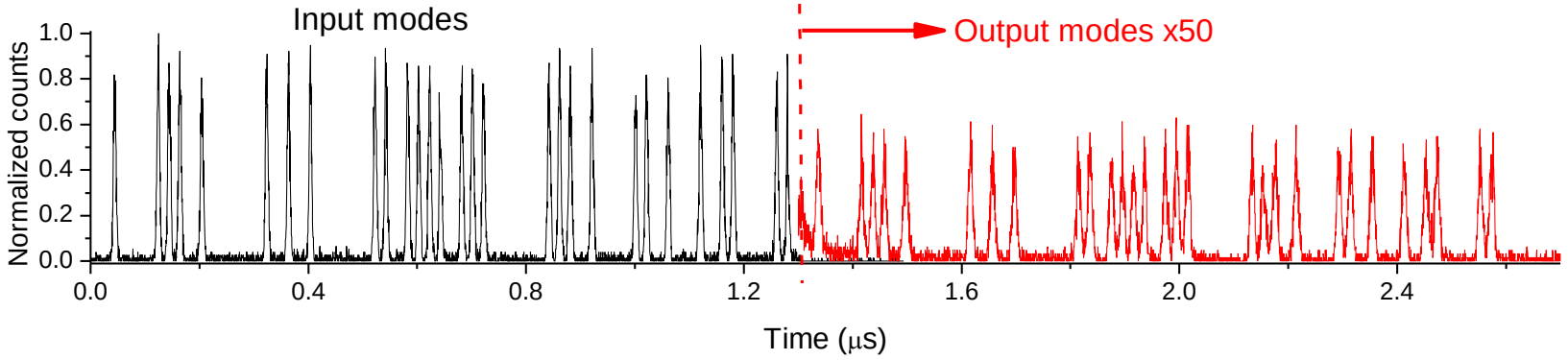
Mapping 64 input modes onto *one* crystal





$$T = 1/\Delta$$
$$= 1.3\mu s$$

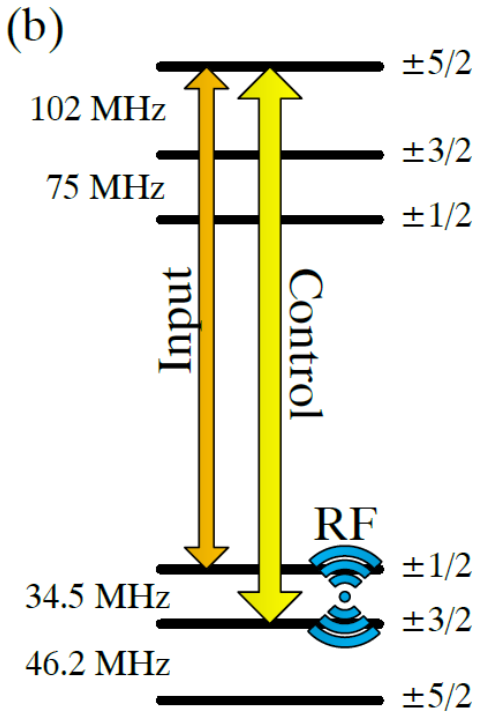
η=1%

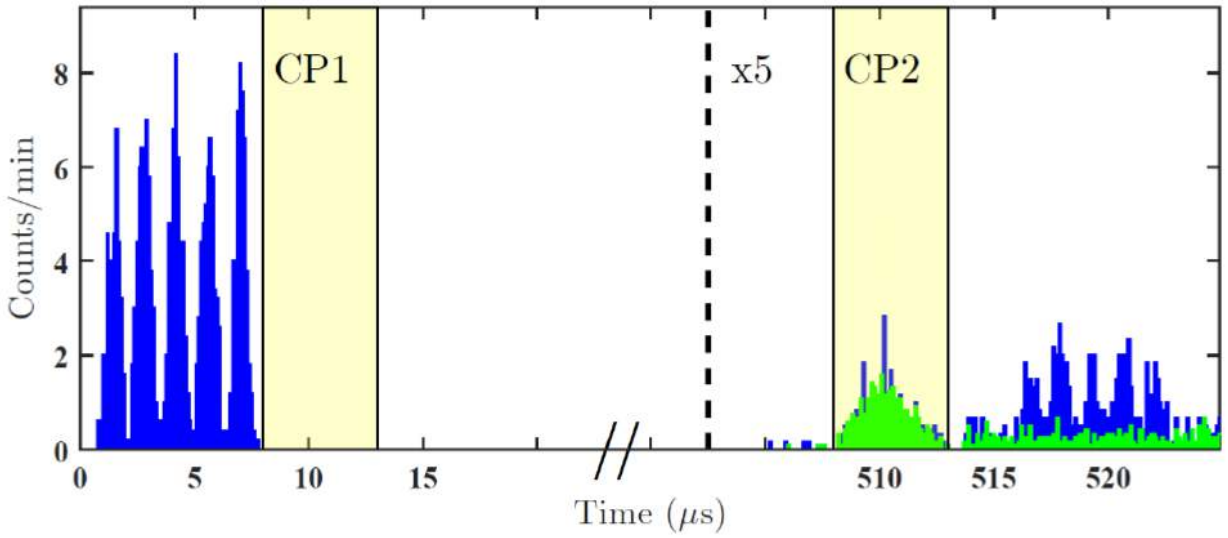


I. Usmani, M. Afzelius, H de Riedmatten and N.Gisin, Nature Communications 1, 12 (2010)

Spin-wave temporal multiplexing

Spin-wave storage of temporally multiplexed polarization qubits in a Eu doped crystal (on demand read-out)





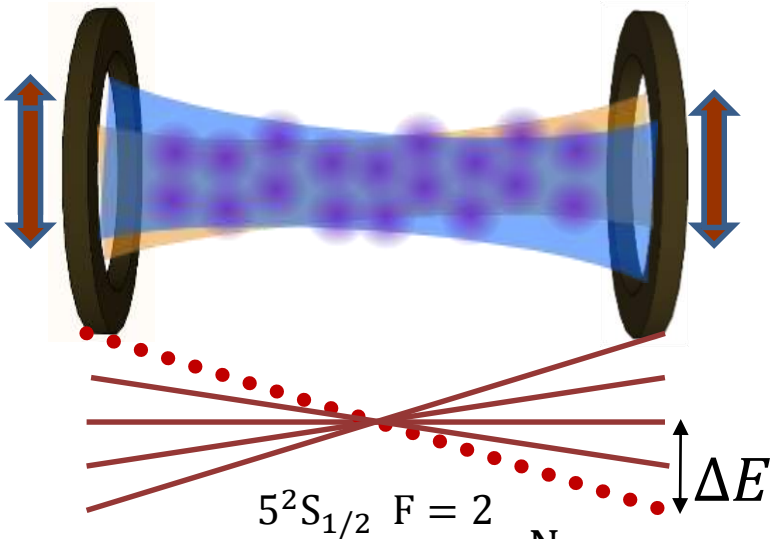
100 modes possible

Eu :YSO

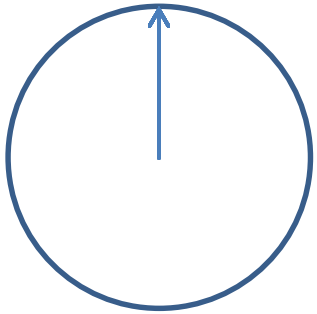
C. Laplane, P. Jobez, J. Etesse, N. Timoney, N. Gisin, M. Afzelius, arXiv:1509.03537
M. Gündogan *et al*, New. J. Phys 045012 (2013)

Scheme for temporally multimode DLCZ memory

- Controlled and reversible inhomogeneous broadening (CRIB) allows the creation of spin waves in multiple temporal modes in a single ensemble
- Broadening of the spin states with magnetic gradients→ different energy separation for each atom between the ground and storage levels → dephasing
- Reversal of the inhomogeneous dephasing → rephasing of the spin waves



Spin wave dephasing

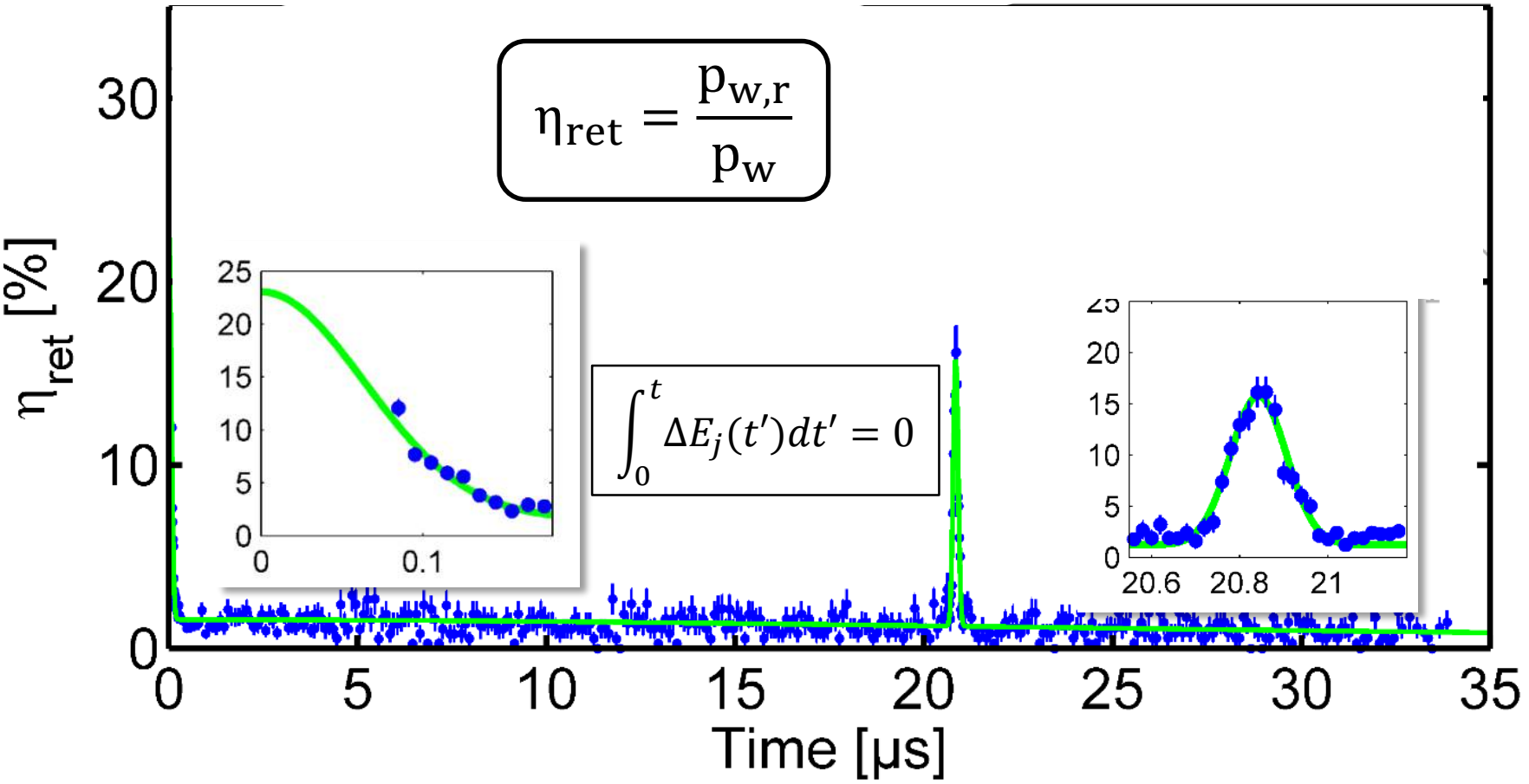


$$|\Psi(t)\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{i\int_0^t \Delta E_j(t')dt'/\hbar} e^{i(\overrightarrow{k_W} - \overrightarrow{k_w})\overrightarrow{x_j}} |g_1 \dots s_j \dots g_N\rangle$$

C.Simon, H. de Riedmatten and M.Afzelius, Phys.Rev.A **82**, 010304 (R)
Temporally multiplexed quantum repeaters in atomic gases

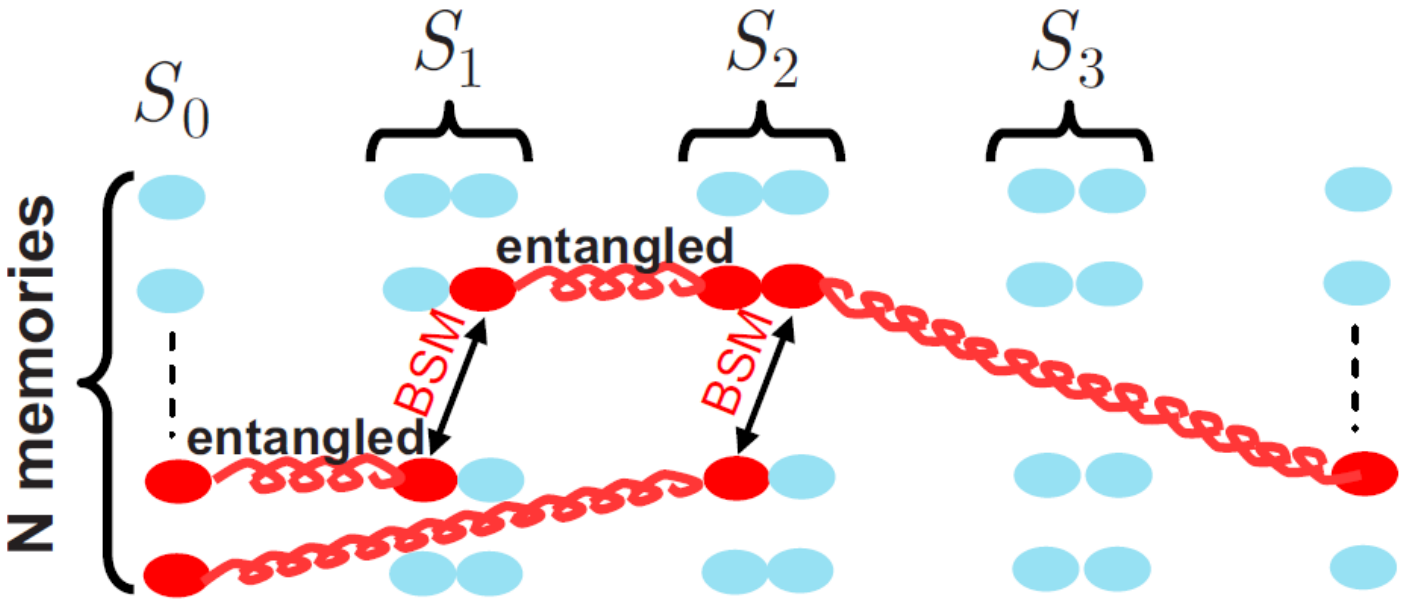
Spin-wave controlled dephasing and rephasing of single collective spin excitations

MAGNETIC FIELD GRADIENT + INVERSION



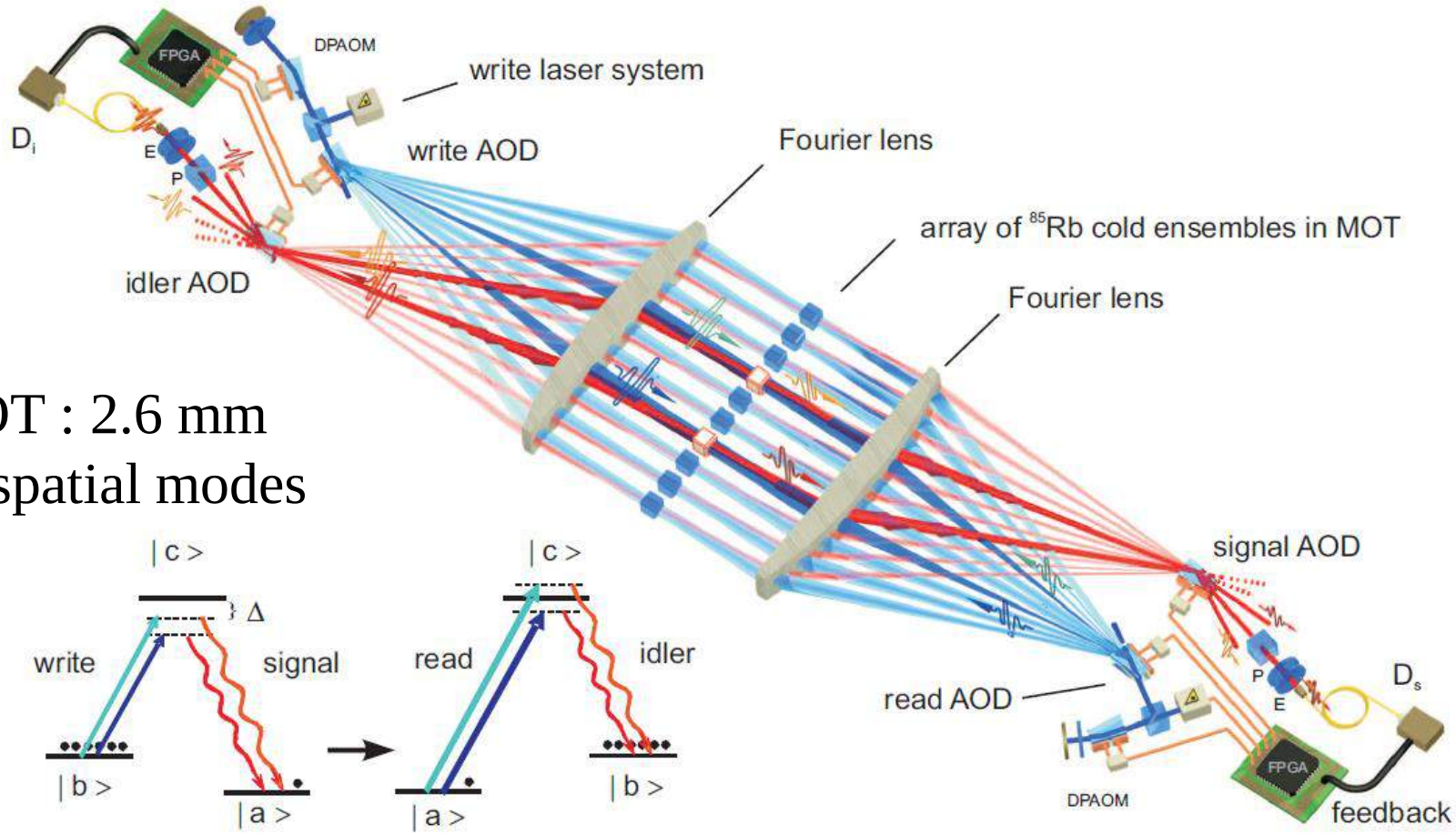
B. Albrecht, P. Farrera, G. Heinze, M. Cristiani, H. de Riedmatten, PRL **115,** 160501 (2015)

Spatially multiplexed quantum repeaters



O. A. Collins, S. D. Jenkins, A. Kuzmich, and T. A. B. Kennedy, Phys. Rev. Lett. 98, 060502 (2007)
M. Razavi, M. Piani, N. Lütkenhaus, Phys.Rev. A 80, 032301 (2009)

Experimental spatially multiplexed quantum memory



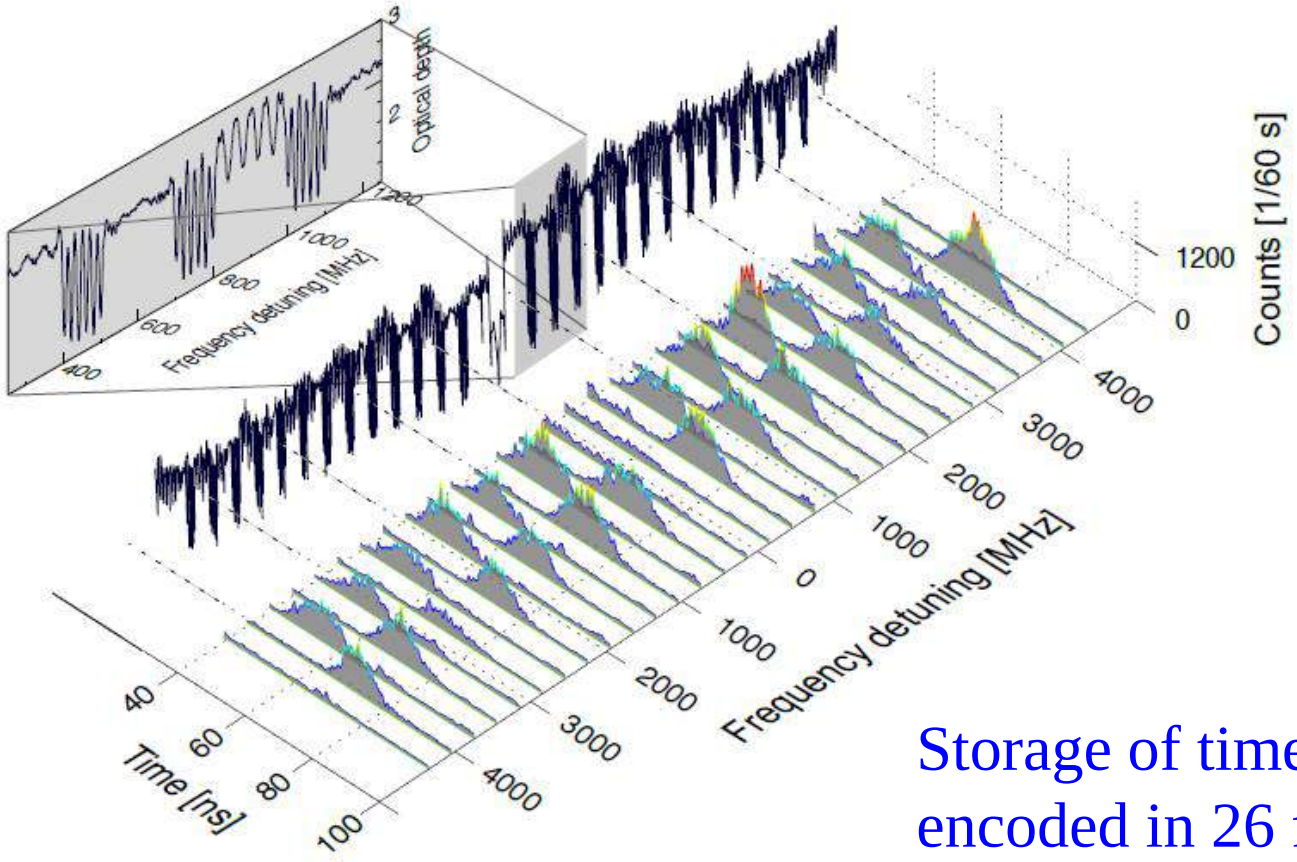
MOT : 2.6 mm
10 spatial modes

100 modes possible with 2D addressing

S.-Y. Lan, A. G. Radnaev, O. A. Collins, D. N. Matsukevich, T. A. B. Kennedy, and A. Kuzmich , Opt. Express, 17, 13639 (2009)

Frequency multiplexing

Take advantage of inhomogeneous broadening in rare-earth doped crystals to create several AFCs at different frequencies

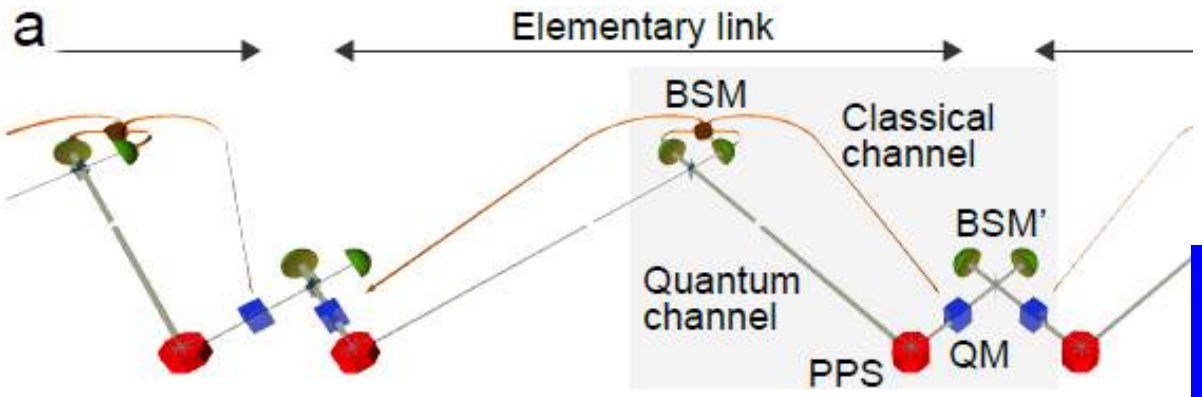


Storage of time bin-qubits encoded in 26 frequency bins
Fidelity 97 %

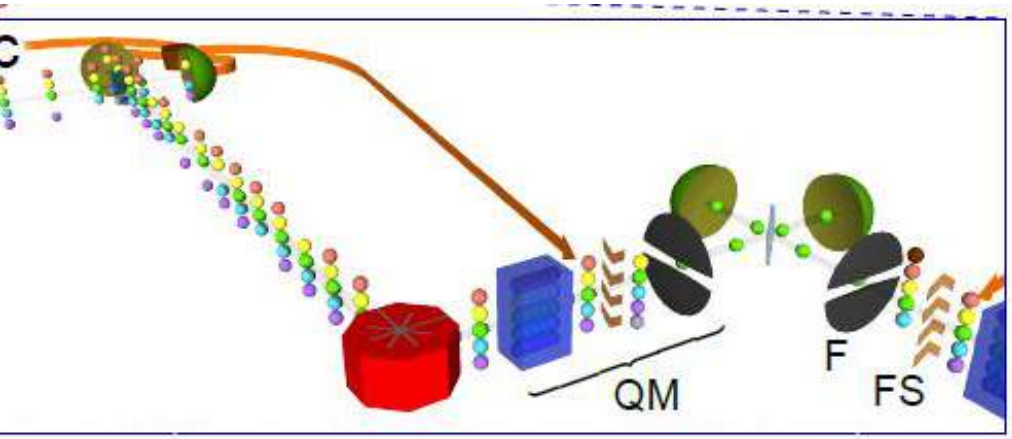
N. Sinclair et al, Phys.Rev.Lett. Phys. Rev. Lett. 113, 053603 (2014) (Tittel group, Calgary)

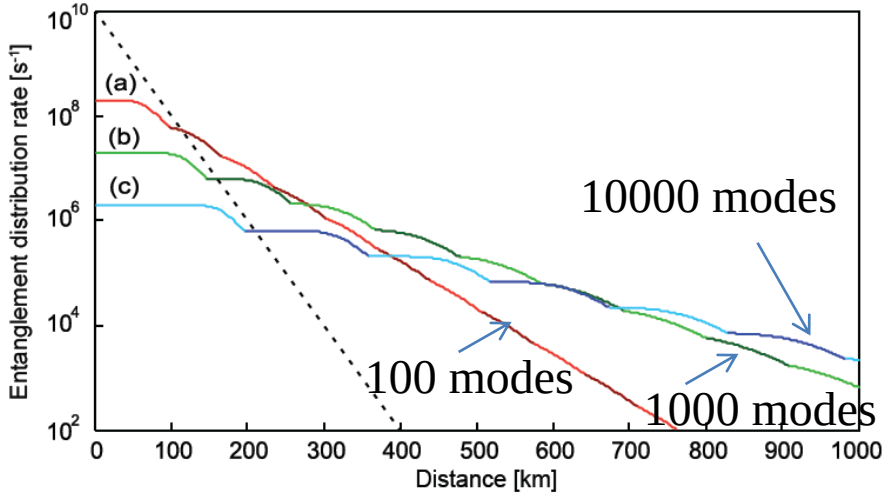
Frequency multiplexing

Selective read-out and feed-forward



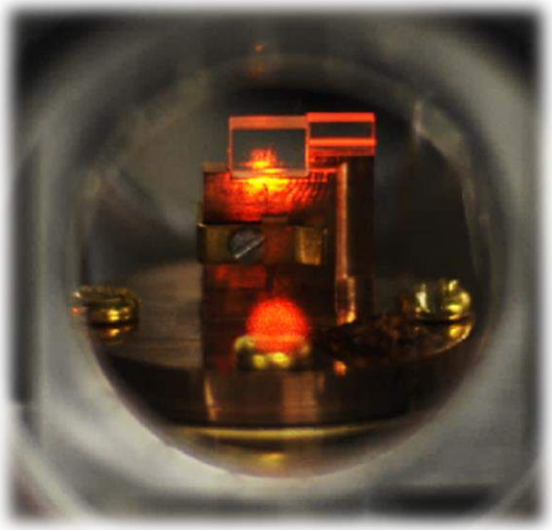
Deterministic sources
QM Bandwidth 300 GHz
Combined spectral and temporal multiplexing





N. Sinclair et al, Phys.Rev.Lett. Phys. Rev. Lett. 113, 053603 (2014) (Tittel group, Calgary)

The dream multimode quantum memory



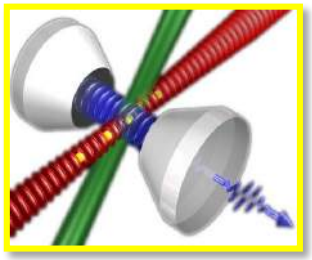
100 temporal modes
100 frequency modes
100 spatial modes

Total : 10^6 modes

Regime of multiple successful entanglement generations
per communication time in elementary segments (not well studied)

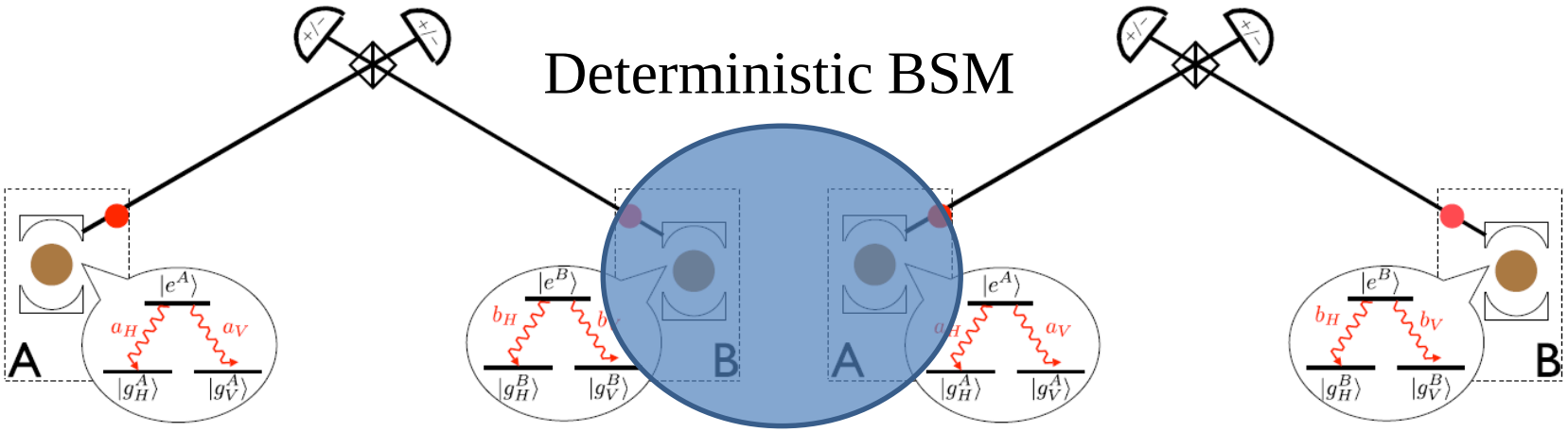
Outline

- Introduction to quantum repeaters
- Quantum repeaters based on atomic ensembles and linear optics
  - The DLCZ protocol
  - Coupling cold atomic QMs to telecom wavelengths
  - Photon pair source and solid-state quantum memories
  - More deterministic schemes
- Multiplexed quantum repeaters
- Quantum repeaters based on single emitters
  - Quantum repeaters based on trapped ions
  - Deterministic atom-photon entanglement
  - Complete BSM



MPQ

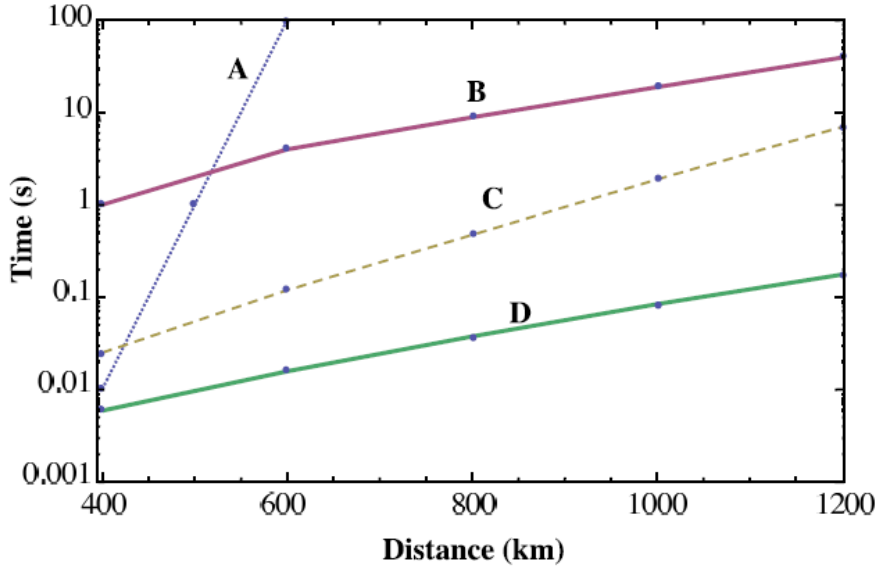
Quantum repeaters based on single trapped ions



$$|\psi_+^{AB}\rangle = \frac{1}{\sqrt{2}}\left(|g_H^A g_H^B\rangle + |g_V^A g_V^B\rangle\right)$$

Entanglement experiments done with NV centers (without cavity) separated by >1 km

Hensen *et al*, arXiv:1508.05949



N. Sangouard, R. Dubessy and C. Simon, Phys.Rev.A 79, 042340 (2009)

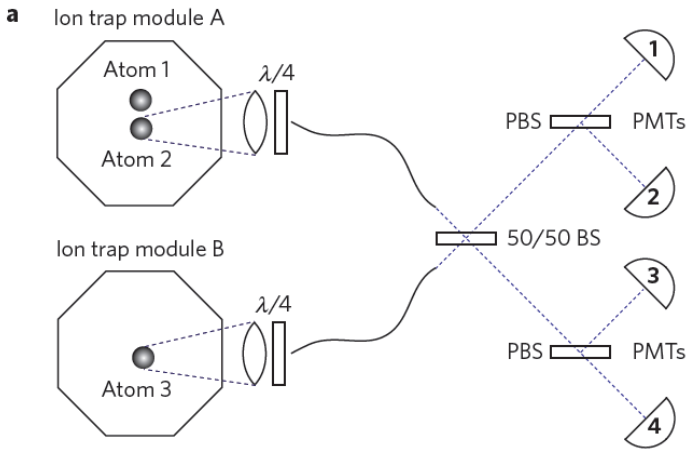
nature physics

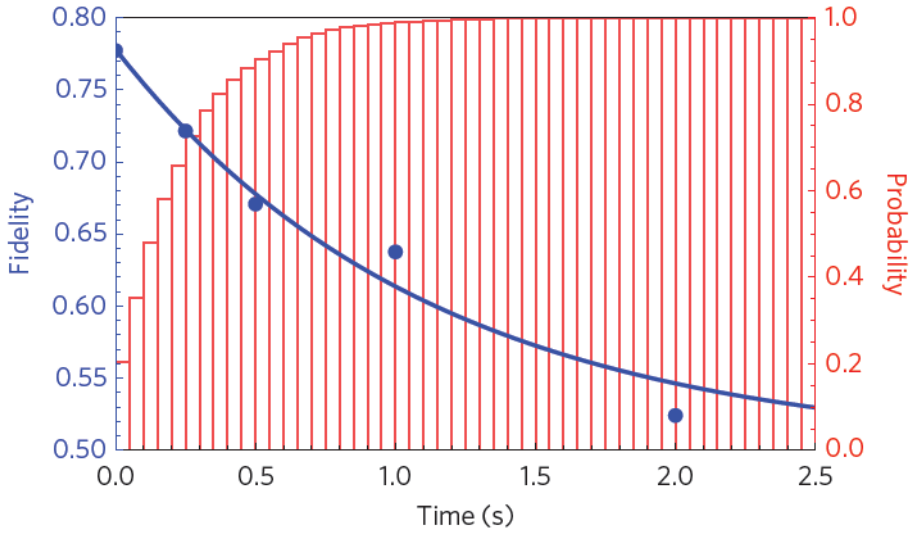
LETTERS

PUBLISHED ONLINE: 17 NOVEMBER 2014 | DOI: 10.1038/NPHYS3150

Modular entanglement of atomic qubits using photons and phonons

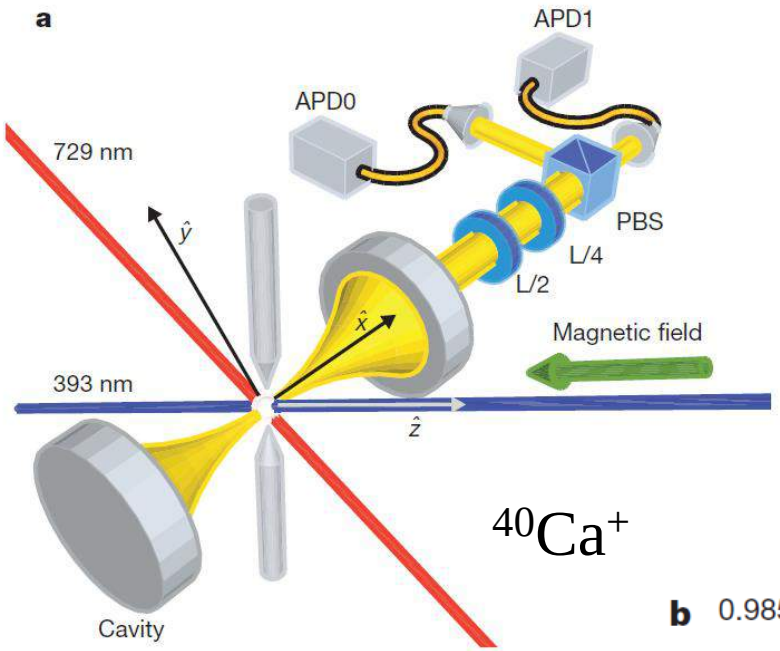
D. Hucul*, I. V. Inlek, G. Vittorini, C. Crocker, S. Debnath, S. M. Clark† and C. Monroe



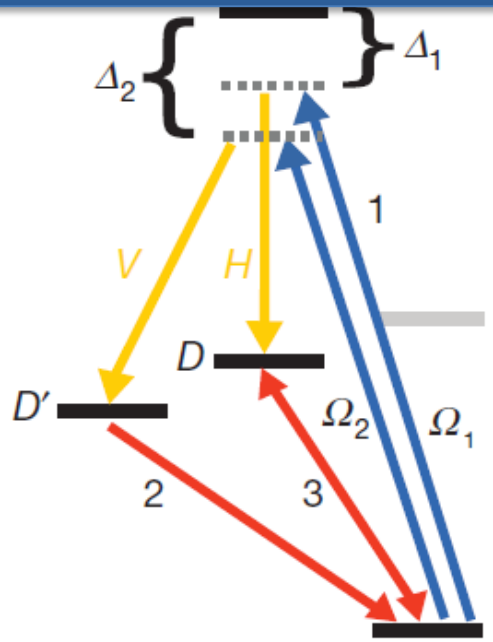


Storage time longer than preparation time

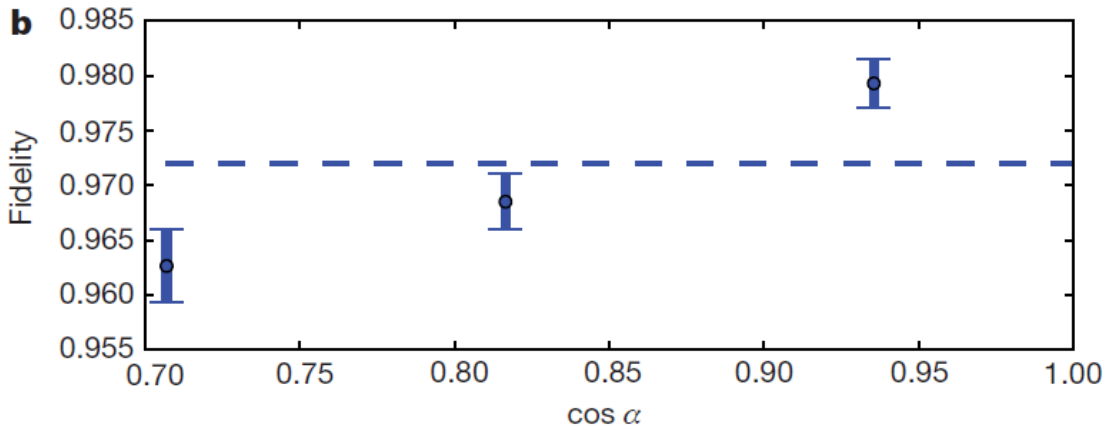
Tunable ion-photon entanglement in an optical cavity



$^{40}Ca^{+}$



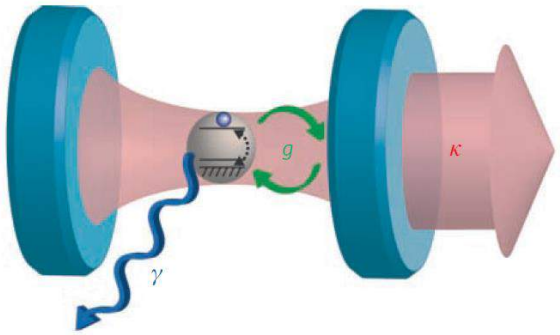
$$|\psi\rangle = \cos\alpha|DH\rangle + e^{i\varphi}\sin\alpha|D'V\rangle$$



Cavity finesse 77000
16 % efficiency

A. Stute, B. Casabone, P. Schindler, T. Monz, P. O. Schmidt, B. Brandstätter, T. E. Northup & R. Blatt
Nature 485, 482 (2012)

High-Fidelity quantum gates with two atoms in cavities



Cooperativity

$$C = \frac{g^2}{\kappa\gamma}$$

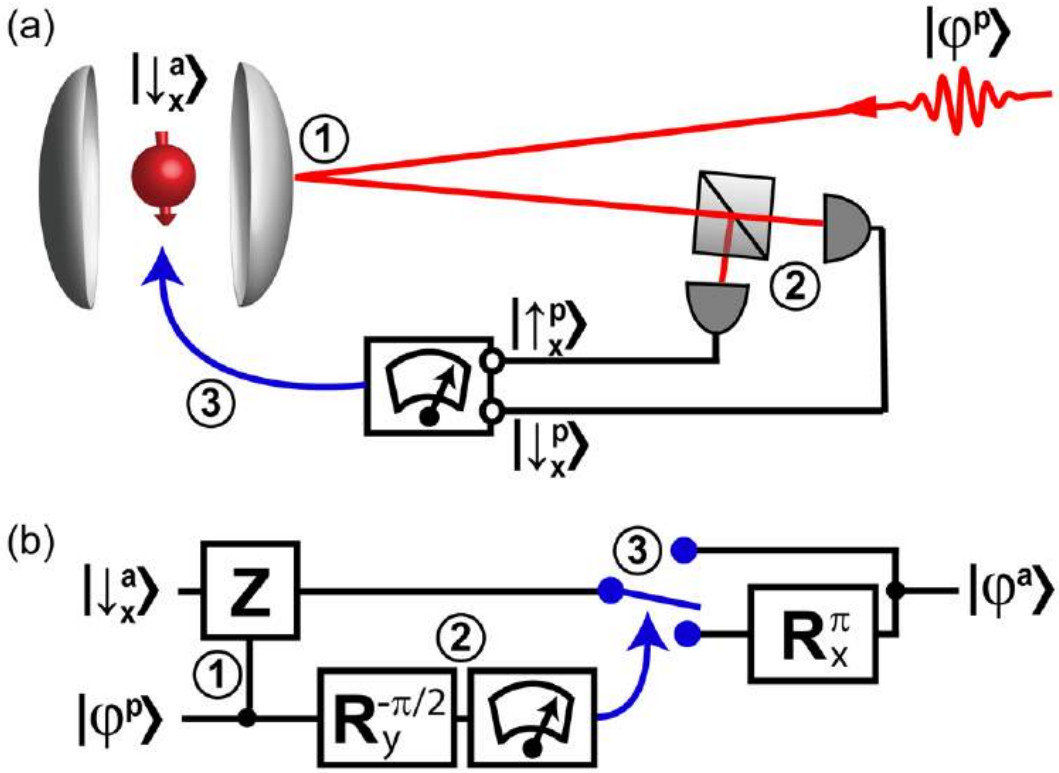
Direct implementation $$F = 1 - \frac{1}{\sqrt{C}}$$

More sophisticated $$F = 1 - \frac{1}{C}$$

A. S. Sørensen and K. Mølmer, Phys. Rev. Lett. 91, 097905 (2003)

J. Borregaard, P. Kómár, E. M. Kessler, M. D. Lukin, and A. S. Sørensen, Phys. Rev. A 92, 012307 (2015)

Heralded single atom quantum memory



Efficiency :39 %
Fidelity : 86 %

N. Kalb, A. Reiserer, S. Ritter, and G. Rempe, Phys. Rev. Lett. 114, 220501 (2015)

Summary

- Atomic ensembles
  - ☺ Simple (strong coupling without high finesse cavities
  - ☺ Multiplexing
  - × Probabilistic atom light entanglement
  - × Probabilistic BSM
- Single atoms
  - ☺ Deterministic atom-light entanglement
  - ☺ Deterministic BSM
  - × Need cavities
  - × no multiplexing

Hybrid repeaters with atomic ensembles for efficient entanglement generation and single atoms (or Rydberg atoms) for deterministic BSM ?

Conclusions

- Distribution of entanglement over continental distance is a great challenge
- Need to combine several technologies
- First goal : beat direct transmission
- New opportunities for QKD
- On the way : fascinating physics and increased quantum control of light-matter interaction.

Acknowledgements

ICFO Institut de Ciències Fotòniques

*iCrea INSTITUCIÓ CATALANA DE RECERCA I ESTUDIS AVANÇATS

GOBIERNO DE ESPAÑA MINISTERIO DE ECONOMÍA Y COMPETITIVIDAD

erc

MARIE CURIE ACTIONS

Marie Curie Initial Training Network
Coherent Information Processing in Rare-Earth Ion Doped Solids



Margherita Mazzera



Andreas Lenhard



HdR



Georg Heinze



David Paredes



Mustafa Gündoğan



Daniel Rieländer



Boris Albrecht



Emanuele Distante



Kutlu Kutluer



Alessandro Seri



Nicolas Maring



Pau Ferrera



Auxiliadora Padron